

PNMF Rationale, Methodology, and Verification & Validation Report

Parametric Noise Modeling Framework (PNMF)
Academic engineering assurance report
Issue date: 2026-08-13

Document control

Table 1. Document control.

Field	Controlled value
Document title	PNMF Rationale, Methodology, and Verification & Validation Report
Document status	Academic assurance baseline for the evidence frozen on 2026-08-13
Intended audience	Aircraft-noise researchers, conceptual-aircraft designers, software reviewers, and academic assessors
Software revision	Git commit <code>595a25979ed3104ff17f698fbd73488addf6cd8b</code> ; worktree <code>dirty=True</code> at validation time [E5, E6]
Canonical datastore	SHA-256 <code>9b3ea2f58347ebb348128c49b3238e6a0b28852858fa7e87c89fad51d5e9fe8e</code> [E5, E6]
Source manifest	SHA-256 <code>6d14674d0cd46c75cc6cc969b4b0ab22bfc885ad1cc09e3ea2560832b78cf510</code> [E19]
Validation seed	<code>20260724</code> [E5, E6]
Configuration status	Verified-scope results are configuration-specific; the dirty worktree prevents treating the commit identifier alone as a complete released baseline [E19]
Approval implication	Technical evidence summary only. This document is not a certification approval, regulatory compliance finding, or authority to use PNMf results for operational limits.

Revision record

Table 2. Revision record.

Revision	Date	Description
1.0	2026-08-13	First evidence-bounded rationale, methodology, verification, and validation report for the current two-route PNMf architecture.

Evidence notation and traceability

Primary literature and official data sources are cited numerically as [1] to [9]. Repository and run evidence is cited as [E1] to [E20]. A code pointer such as `pnmf/models.py:90-99` is relative to `projects/pnmf/` and identifies the implementation reviewed for this report. Generated historical prose is not used as an external scientific authority.

Table 3. Local evidence register.

ID	Evidence item	Scope
E1	<code>pnmf/anp.py:21-23, 41-71, 135-206, 294-310, 339-400, 409-564, 625-646</code>	Paths, input tables, provenance, collision policy, manifest, and truth/prediction separation
E2	<code>pnmf/core.py:53-76, 270-324, 334-342</code>	Aircraft descriptors, standard NPD distances, and baseline lookup
E3	<code>pnmf/models.py:37-73, 76-140, 162-198</code>	Learned features, power transformations, ET/RF definitions, multi-output fit, monotone projection, and tree dispersion
E4	<code>pnmf/api.py:37-48, 150-171, 187-264</code>	Public learned surface, power-grid contract, prediction metadata, and versioned physics-artifact loader
E5	<code>tmp/reason-report/model-validation-current/run_manifest.json, summary.csv, and MODEL_TRAINING_REPORT_CURRENT.md</code>	Fresh grouped and temporal learned validation, environment, hashes, metrics, counts, and limitations
E6	<code>tmp/reason-report/jet-reference-current/run_manifest.json, summary.csv, predictions.csv, and JET_REFERENCE_VALIDATION_REPORT_CURRENT.md</code>	Frozen v6.3 jet holdout, reference selection, train/test separation, metrics, and figures
E7	<code>pnmf/physics.py:30-83, 125-326, 467-735, 741-769, 785-1002, 1004-1025, 1087-1315</code>	Physical input contracts, component formulas, spectra, event calculation, exclusions, output scope, and calibration load/apply hooks
E8	<code>tmp/reason-report/physics-run/outputs/physics_fleet_validation.csv</code> and <code>physics_fleet_validation.png</code>	Dated pre-scope physics comparator retained for historical context
E9	<code>tests/test_migration.py:16-67</code>	Corpus counts, mandatory v6.3 behavior, supported learners, and caller-CWD independence

ID	Evidence item	Scope
E10	tests/test_pnmf.py:69–115	Doc 29-style interpolation, extrapolation, and power-unit transformation tests
E11	tests/test_validation.py:12–44, 47–96, 99–135	Group leakage control, metric formulas, hyperparameters, and artifact contract
E12	tests/test_jet_reference_validation.py:24–91, 94–128, 131–186	Target-blind reference selection, exact split contract, correlated-cell counts, and result artifact contract
E13	tests/test_physics.py:18–107, 123–193, 196–284	Propagation, source gating, metrics, typed inputs, six airframe components, energy conservation, and diagnostics
E14	Historical tmp/reason-report/assurance_execution_record.md, SHA-256 7b8e908e485bba674c1200a7eaa1bfff23e7a092784c7dd1557088c1a5fd9c9eb; final rerun tmp/reason-report/verified_scope_20260813/full_suite_result_final3.txt, SHA-256 e141c847f18a19aef910cd64dce4e1430f2e3680703eb0771712ccceca9d116a	Dated commands and observed results for the manifest, historical bounded attempt, current terminating full suite, focused tests, generated validation, and physics run
E15	tmp/reason-report/fresh_vs_dated_comparison.json, SHA-256 a5d71361e29e460af3b30ccb7834a5bfead9b4db3c345abb9413d5f37eecdabc2	Input paths/hashes, comparison keys, row-count agreement, and maximum numeric differences for learned and jet summaries
E16	pnmf/validation.py:37–67, 150–190, 240–330, 823–960, 1000–1055	Validation combinations, sample construction, grouped/temporal split logic, report wording, and run manifest
E17	pnmf_cli.py:287–430	Physics calibration/fleet-validation workflow and plot generation
E18	tmp/reason-report/primary_source_notes.md	Primary-source cross-check, visually checked pages, and citation boundary
E19	tmp/reason-report/verified_scope_20260813/run_manifest.json, summary.csv, verified_vs_merged_delta.csv, physics_validation.csv, and physics_calibration_used.json	Fresh verified/merged grouped, temporal, frozen-jet, and physics evidence with seed, counts, hashes, and provenance
E20	pnmf/verified_anp.py, pnmf/physics_presets.py, pnmf/physics_calibration.py, pnmf/physics_calibration_A320–270N_v1.json, and focused tests	Versioned verified registry, fail-closed resolver, eleven physics presets, and serialized calibration contract

Abstract and executive assurance conclusion

PNMF exists to generate and compare aircraft noise-power-distance (NPD) tables during conceptual design when a directly measured or manufacturer-supplied NPD set is unavailable. It combines two deliberately independent routes. The learned route fits Extra Trees (ET) or Random Forest (RF) regressors to provenance-controlled EASA Aircraft Noise and Performance (ANP) data. The component route predicts idealised SEL and LAmass events from typed engine, airframe, atmosphere, and trajectory inputs using reduced physically motivated source laws. Both routes emit the same NPD table shape, allowing engineering comparison without allowing either route to train or correct the other [E3, E4, E7].

The assurance conclusion is mixed by design. Software and data-contract claims are the strongest part of the case: source provenance is embedded; legacy v2.3 and v6.3 data are merged by declared business keys; ANP truth and generated predictions occupy separate tables; supported learners and hyperparameters are fixed; grouped split construction prevents exact aircraft/NPD leakage; physics supports only SEL and LAmass; and unit/provenance contracts are tested [E1, E3, E7, E9–E13]. These claims are **Verified** for the reviewed revision and evidence bundle.

Scientific predictive validity is weaker. The fresh verified run uses exactly eleven trainable v6.3 families, three aircraft-grouped folds, and the fixed seed **20260724**. It evaluates verified-only fits and merged comparators on identical held-out folds, a legacy-to-v6.3 temporal protocol for the merged comparator, and a frozen three-jet reference holdout. The verified sample contains 420 power rows across eleven NPD IDs; the resulting power-distance cells are correlated observations, not independent aircraft [E19]. The serialized A320–270N physics calibration is an in-sample fit over 180 SEL/LAmass cells with 6.20 dB RMSE; the remaining ten verified families are evaluated with that artifact held fixed [E19, E20]. These results are **Empirically supported within scope**, not universal validation.

The evidence does not establish predictive accuracy for genuinely novel aircraft families, unconventional architectures, prospectively acquired data, component source spectra, certified noise levels, or a complete Doc 29 event/contour implementation. It does not establish calibrated uncertainty intervals. The current full software suite terminates with 125 passed and 1 skipped test; this verifies the reviewed software test contract but does not establish scientific accuracy or certification suitability [E14]. Those latter claims are **Not established**.

The defensible use is therefore conceptual screening: generate traceable NPD-like estimates, inspect sensitivity and route disagreement, and identify where better inputs or external evidence are required. PNMf output must not be represented as aircraft certification evidence, a regulatory noise contour, a measured NPD curve, or a full ANOPP, Stone, Heidmann, Fink, or ECAC Doc 29 reproduction.

1. Why PNMf exists

Aircraft concepts are routinely evaluated before a production engine deck, measured flyover data, or an official ANP entry exists. At that stage, an engineer may still need a structured estimate of how single-event level changes with power, operating mode, and slant distance. Directly copying a nearby aircraft curve obscures the relationship between design descriptors and the result. A pure component model, meanwhile, may require detailed geometry and thermodynamic states that are not yet available. PNMf addresses this gap by offering two complementary, inspectable estimates rather than claiming one complete predictive truth.

The learned route answers: given the descriptors and requested power rows for a proposed aircraft, what ten-distance NPD shape is statistically consistent with the staged ANP population? The physics route answers: given component-relevant quantities and explicit assumptions, what SEL or LAmass event emerges from the reduced source and propagation calculation? Their agreement may increase confidence that an estimate is directionally plausible; disagreement is a review signal. Agreement cannot prove truth because the routes share broad physical context and the physics calibration ultimately uses ANP truth for one aircraft.

EASA describes ANP as a resource collected, verified, and made available for airport-noise modelling under its Regulation (EU) No 598/2014 mandate [3, 4]. ECAC Doc 29 explains that long-term average noise exposure requires a method, suitable ANP data, and operations/scenario data; its methodology is not intended to predict the absolute level of an individual movement reliably [1, Sections 1.1-1.2]. PNMf occupies an earlier and narrower layer. It generates NPD tables or source-side NPD-like tables for screening. It does not execute the entire segment, installation, lateral attenuation, terrain, ground, atmosphere, and contour chain described in Doc 29 [1, Sections 4.2-4.5].

The value proposition is therefore traceability under incomplete information:

1. preserve the provenance of every ANP row used for training;
2. expose the exact learned feature and power contracts;
3. preserve independent component reasoning and physical input status;
4. return standard, reviewable NPD tables rather than an opaque scalar score;
5. quantify retrospective residuals without converting them into certification claims; and
6. make exclusions and missing evidence explicit.

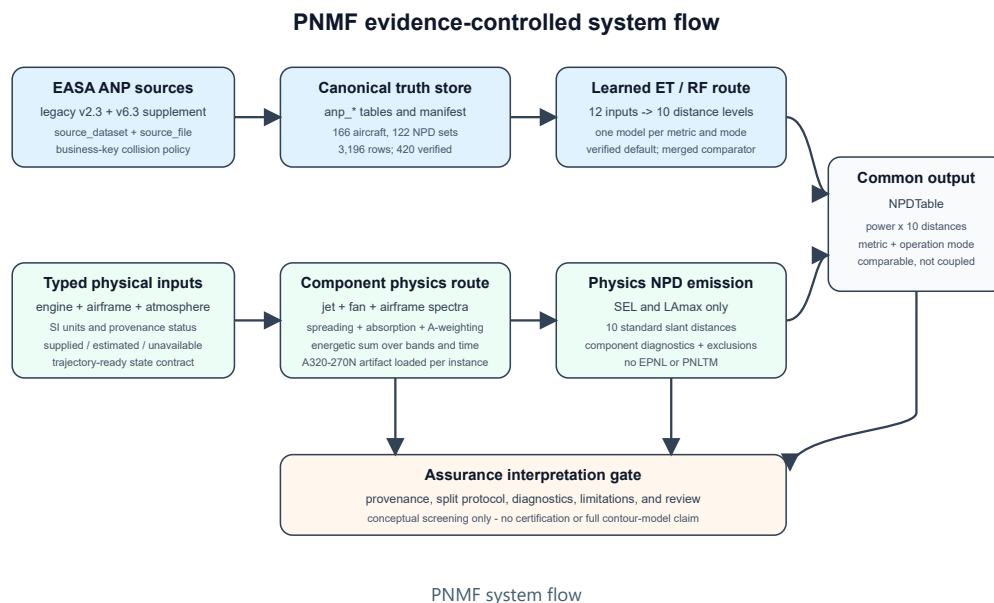


Figure 1. PNMf evidence-controlled system flow. Source: report-specific synthesis of `pnmf/anp.py`, `pnmf/models.py`, `pnmf/api.py`, and `pnmf/physics.py` [E1-E4, E7].

2. Applicability and non-certification boundary

2.1 Intended applications

PNMF is appropriate for comparative concept screening when the user records the input provenance, stays near represented technologies, retains both route outputs as estimates, and subjects material decisions to engineering review. Typical questions include relative changes with installed thrust, bypass ratio, engine count, weight, airframe configuration, or requested power grid; construction of a provisional NPD table for a conceptual workflow; and identification of missing component inputs that dominate model uncertainty.

It may also support software-method research, for example comparing grouping strategies, studying release transfer, or testing how a physical-shape constraint affects learned curves. Such work remains retrospective unless a dataset is withheld prospectively and isolated from model-selection feedback.

2.2 Explicit non-applications

PNMF output is not suitable, by itself, for:

- aircraft noise certification under ICAO Annex 16 or an equivalent national rule;
- a compliance finding against an airport, land-use, or operating restriction;
- a regulatory airport noise contour;
- absolute prediction of a specific measured movement;
- replacement of a manufacturer-supplied or approved ANP entry;
- approval of a novel engine or airframe architecture outside the evidence population;
- calibrated probabilistic risk or confidence limits; or
- claims of full ECAC Doc 29, ANOPP, Stone, Heidmann, or Fink implementation.

This boundary follows both source scope and implementation scope. Doc 29 Volume 3 provides fixed verification cases for establishing implementation equivalence to Volume 2 and distinguishes that verification from a future measurement-validation framework [2, PDF pp. 10-16]. PNMf has not produced those Volume 3 equivalence artifacts [E18]. The physics implementation explicitly excludes installation/shielding,

nacelle treatment, ground reflection, lateral attenuation, terrain, and non-uniform atmosphere, and it rejects EPNL and PNLTm [E7]. Those are not minor presentation omissions; they prevent a complete event/contour or certification claim.

2.3 Safe result language

The preferred statement is:

PNMF applies the ECAC Doc 29 baseline NPD interpolation convention to EASA ANP-derived learned estimates and provides a separate, A320-270N-anchored component-physics cross-check for SEL and L_{Amax}. The outputs are conceptual screening evidence and are not certification or full contour-model results.

Avoid language such as "validated for all aircraft," "Doc 29 compliant," "ANOPP equivalent," "certified accuracy," or "confidence interval" unless future evidence directly supports the specific statement.

3. System context and architecture

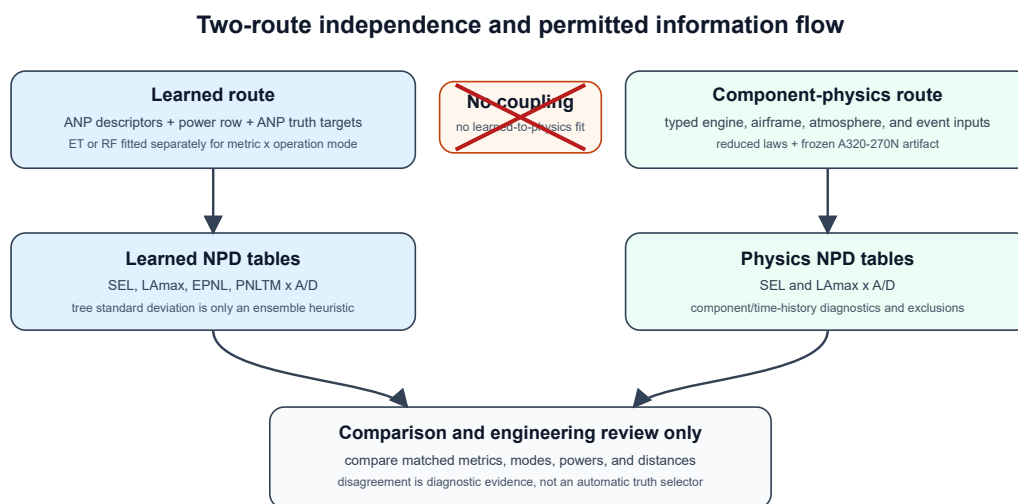
3.1 System boundary

The PNMf application boundary contains raw-source ingestion, a canonical SQLite datastore, learned model fitting/prediction, an independent component-physics calculation, NPD table lookup, validation commands, command-line and Streamlit interfaces, and guarded prediction storage. External to the boundary are EASA's source stewardship, certification measurements, manufacturer design data, downstream airport scenario modelling, and regulatory decision processes.

The central interchange object is **NPDTable**: power settings by ten standard slant distances, with a noise metric and approach (A) or departure (D) mode. The standard distances are 200, 400, 630, 1,000, 2,000, 4,000, 6,300, 10,000, 16,000, and 25,000 ft [E2]. The learned route can emit SEL, L_{Amax}, EPNL, and PNLTm for both modes. The physics route can emit only SEL and L_{Amax} for both modes [E3, E7].

3.2 Independence claim

ET and RF are fitted only from descriptor/power inputs and ANP noise targets. The physics model receives no learned prediction, tree residual, tree dispersion, fitted feature importance, or learned correction. Its five calibration variables are fitted only to A320-270N ANP SEL/L_{Amax} targets, after which the versioned artifact is loaded and reused for physics predictions [E3, E4, E7, E20]. A comparison layer can compare matched outputs. It must not feed that comparison back into physics automatically.



Permitted: output-to-output comparison. Prohibited: learned predictions or residuals fitting, correcting, or driving component physics.

Two-route independence

Figure 2. Learned and component-physics route independence. Source: report-specific architecture trace from `pnmf/models.py:102-140`, `pnmf/api.py:150-164`, `237-243`, and `pnmf/physics.py:1226-1286` [E3, E4, E7].

The term "independent" needs qualification. It means computational and fitting independence in PNMf. It does not mean independent external evidence: both routes reflect conventional aeroacoustic knowledge, and the physics anchor is one ANP aircraft. Consequently, agreement is a consistency check rather than a statistically independent confirmation of truth.

3.3 Operational flow

For the learned path, the canonical datastore is opened relative to the application rather than the caller's working directory. The selected learner is fitted for each available metric/mode pair. The user supplies a **ParametricAircraft** and either a common power sequence or separate approach/departure sequences. Each grid is canonicalised to positive, finite, unique, ascending lbf per engine. The model returns an NPD table and optional tree dispersion [E4].

For the physics path, the user defines a **PhysicsDesign** with required summary quantities and optional typed component inputs. The calculator builds an engine state, applies the mode configuration, evaluates available source components at a sequence of emission angles, propagates one-third-octave spectra, applies A-weighting, and integrates the event. **NoisePredictor** loads and validates the versioned A320-270N calibration artifact only when the physics route is requested, then caches the loaded object in the same predictor instance [E4, E7, E20]. Runtime never silently refits or falls back to A320-211.

3.4 Verified registry and training scopes

The verified-data boundary is represented by the versioned registry `easa-verified-14-family-v1` [E20]. It contains the fourteen official-list families supplied by the reviewed EASA workbook: eleven `trainable` families with complete local v6.3 numerical tables and three `verified_metadata_only` families, A320-250N, A330-941, and G650ER. Metadata-only verification is displayed as unavailable and does not create numerical training truth.

`TrainingScope` is the explicit selector in the model/API surface. `verified` is the default and resolves exactly the eleven named v6.3 NPD IDs after checking aircraft/NPD provenance, all eight metric/mode tasks, and the pinned workbook, aircraft, and NPD hashes. `merged` retains the historical legacy-plus-supplement corpus as a labelled research comparator; it does not mean that the supplement is unverified. Future supplement rows are not admitted automatically: a registry review and version change are required [E20].

The resolver fails closed when a trainable registry member is absent, has non-v6.3 provenance, loses any required task, or is excluded by an incomplete source contract. Model metadata records learner, scope, registry version, selected NPD IDs, support counts, and source hashes. Prediction storage uses `et-verified`, `rf-verified`, `et-merged`, or `rf-merged` identities without altering the truth datastore schema [E20].

4. Data and input contracts

4.1 Canonical ANP data contract

PNMF ingests the staged EASA ANP legacy v2.3 directory and the v6.3 supplement. The legacy files use semicolon delimiters and the supplement uses commas. Three logical tables are mandatory: aircraft, NPD data, and jet-engine coefficients. Optional tables cover propeller coefficients, aerodynamics, procedures, profiles, and weights [E1].

Every imported row receives `source_dataset` and `source_file`. The merge verifies schema equality after excluding those provenance columns, then applies table-specific business keys. Because input order is legacy then supplement, `drop_duplicates(..., keep="last")` makes v6.3 win a collision deterministically. Aircraft records use the composite key `ACFT_ID`, `NPD_ID`, preserving both 7773ER rows because they refer to different NPD sets. NPD rows use `NPD_ID`, `Noise Metric`, `Op Mode`, `Power Setting` [E1].

The current manifest contains 166 aircraft records, 122 unique NPD sets, and 3,196 NPD power rows. Of the NPD rows, 2,776 originate from `legacy_v2.3` and 420 from `supplement_v6.3` [E5, E9]. These are datastore population counts, not independent observations for predictive inference: distance cells within a row and metric/mode curves for an aircraft are correlated.

The build fails if required v6.3 inputs are missing or contribute no usable rows. It also checks that aircraft and NPD identifiers are referentially consistent and that v6.3 NPD identifiers retain v6.3 aircraft descriptors [E1, E9]. A manifest table captures logical table, source, file, input row count, merge key, collision policy, combined count, and removed duplicates.

Truth and prediction storage are deliberately separated. Canonical truth occupies `anp_*` tables written only by the datastore builder. Generated results occupy `predicted_aircraft` and `predicted_npd` and pass a finite-value, plausible-range, distance-monotonicity, and power-grid gate before storage [E1]. The plausible-level and caution values in this storage gate are operational QA settings, not predictive validation pass/fail thresholds. This report introduces no arbitrary accuracy acceptance threshold.

4.2 Learned aircraft descriptor contract

One supervised sample corresponds to one NPD power row. Its exact 12-element input vector is:

```
[is_jet,      is_turboprop,   is_piston,    n_engines,    log10(MTOW_lb),    log10(MLW_lb),    MLW/MTOW,
log10(static_thrust_lb_per_engine), log10(total_static_thrust_lb), noise_chapter, log10(power_lb), throttle]
```

[E3, E5].

Table 4. Learned input vector and transformation contract.

Index	Input	Transformation and interpretation	Evidence
1	<code>is_jet</code>	One-hot indicator for engine type Jet	<code>pnmf/core.py:59–65</code> [E2]
2	<code>is_turboprop</code>	One-hot indicator for engine type Turboprop	<code>pnmf/core.py:59–65</code> [E2]
3	<code>is_piston</code>	One-hot indicator for engine type Piston	<code>pnmf/core.py:59–65</code> [E2]
4	<code>n_engines</code>	Numeric engine count, distinct from engine-type encoding	<code>pnmf/core.py:66</code> [E2]
5	<code>log_mtow</code>	<code>log10(MTOW_lb)</code> ; positive input required by the validation sample builder	<code>pnmf/core.py:67</code> ; <code>pnmf/validation.py:150–190</code> [E2, E16]
6	<code>log_mlw</code>	<code>log10(MLW_lb)</code> ; positive input required	<code>pnmf/core.py:68</code> ; <code>pnmf/validation.py:150–190</code> [E2, E16]
7	<code>mlw_mtow</code>	<code>MLW_lb / MTOW_lb</code>	<code>pnmf/core.py:60, 69</code> [E2]
8	<code>log_thrust_per_eng</code>	<code>log10(max(static_thrust_lb_per_engine, 1))</code>	<code>pnmf/core.py:70</code> [E2]
9	<code>log_total_thrust</code>	<code>log10(max(static_thrust_lb_per_engine x n_engines, 1))</code>	<code>pnmf/core.py:61, 71</code> [E2]
10	<code>noise_chapter</code>	Numeric ANP aircraft noise-chapter field	<code>pnmf/core.py:72</code> [E2]
11	<code>log_power_lb</code>	<code>log10(max(unit-corrected power_lb, 1))</code>	<code>pnmf/models.py:41–62</code> [E3]
12	<code>throttle</code>	Unit-dependent ratio, clipped to <code>[0, 2]</code>	<code>pnmf/models.py:41–62</code> [E3]

Power conversion is explicit. For CNT (`lb`), `power_lb=P` and `throttle=P/T_static`. For percent corrected net thrust, `power_lb=(P/100)T_static` and `throttle=P/100`. For RPM, no exact thrust conversion is asserted: PNMf uses `throttle=P/max(P_grid)` and `power_lb=throttle T_static`, which is a documented approximation. Unrecognised power-parameter text falls through to the `lb` interpretation [E3]. That default should be treated as an input-quality risk, not evidence that arbitrary units are physically interchangeable.

The target is a ten-element vector of truth levels at the standard distances. The held-out aircraft descriptor and its power row are valid inputs because they define the requested table. Held-out noise values are used only after prediction for scoring. This distinction is material in release holdout: the temporal model fits legacy noise targets only [E5].

4.3 Learned operation and output request contract

The public interface supports exactly ET (`et`, default) and RF (`rf`). It fits the selected learner independently for each available combination of SEL, LAmass, EPNL, and PNLTM with approach and departure. Default departure powers are four points from 0.45 to 0.95 of static thrust; default approach powers are three points from 0.07 to 0.35 [E4]. Explicit mode-specific grids are preferred when engineering intent is known.

Input acceptance requires finite, positive, unique powers. The result preserves the requested ascending grid and returns one (`P`, `10`) level matrix per metric/mode. The operation mode is categorical; it is not interpolated. Units at this learned boundary are feet for distance and lbf per engine for corrected net thrust unless the training row's source power parameter invokes a documented transformation.

4.4 Component-physics input contract

The physics route uses SI internally. Lbf and feet are converted only at the physics boundary [E7]. `PhysicalInput` couples a value to a status of `supplied`, `estimated`, or `unavailable`, plus a note. A missing or explicitly unavailable value cannot be presented as supplied. This mechanism applies to engine, airframe, atmospheric, and trajectory-ready inputs.

Table 5. Physics input groups.

Group	Contracted quantities	Use and boundary	Evidence
Engine summary	thrust, bypass ratio, engine count, maximum thrust	Minimum conceptual engine state and scaling	<code>pnmf/physics.py:785–805</code> [E7]
Detailed jet streams	velocity, mass flow, diameter, temperature, optional total pressure, stream name	Enables detailed stream path when provided	<code>pnmf/physics.py:156–166, 785–820</code> [E7]
Fan deck	mass flow, tip speed, blade count, rotor diameter, temperature rise, optional RPM/N1/stator/spacing	Enables detailed reduced Heidmann-informed fan path only when the gate is complete	<code>pnmf/physics.py:168–180, 821–845</code> [E7]
Optional core	velocity, mass flow, diameter, temperature, combustor exit temperature, total pressure, turbine attenuation	Core remains unavailable until every required field is complete	<code>pnmf/physics.py:183–193, 847–903</code> [E7]
Typed engine fields	18 provenance-bearing physical values	Preserves source status through adapters	<code>pnmf/physics.py:196–217</code> [E7]
Airframe	wing/span; flap/slat areas, chords, angles; nose/main wheel and strut data; gear state	Feeds reduced wing, slat, flap, and gear components	<code>pnmf/physics.py:219–255, 927–981</code> [E7]
Atmosphere	temperature, relative humidity, pressure	Molecular absorption; defaults are 15 degC, 70 percent RH, and 101.325 kPa	<code>pnmf/physics.py:257–268, 1007–1017</code> [E7]
Trajectory-ready state	position, speed, Mach, altitude, attitude, thrust, configuration	Complete future state contract; missing required fields raise an error	<code>pnmf/physics.py:271–326</code> [E7]

Where explicit geometry is absent, the current `PhysicsDesign` estimates wing area from 600 kg/m² wing loading, span from aspect ratio 9, and wheel count from weight bands [E7]. These values are model assumptions. They must remain labelled estimated and should not be mistaken for aircraft-specific measurements. Similarly, fallback source paths are lower-information conceptual approximations. Detailed engine decks and measured component evidence are preferred.

The reference NPD event is steady level flight at 160 kt with the requested per-engine thrust and closest slant distance. Approach defaults to landing flap and gear down; departure defaults to takeoff flap and gear up [E7]. The typed trajectory contract can carry richer state, but the current NPD adapter remains a reference-flight-path calculation rather than a general segmented airport trajectory.

5. Learned ET/RF methodology

5.1 Model structure

PNMF creates a separate regressor for each noise metric and operation mode (`pnmf/models.py:102–149`). Within that regressor, the ten distance levels are fitted jointly as a native multi-output target. Each tree predicts a ten-element vector; the ensemble prediction is the arithmetic mean across its trees, as defined for the scikit-learn regressors [8, 9] and consumed at `pnmf/models.py:173–190`. Joint output does not impose an acoustic covariance model and does not make the ten cells statistically independent.

Extra Trees randomises split thresholds more strongly than a conventional random forest and averages many trees to reduce variance [8]. Random Forest fits bootstrap-based decision trees and averages them [9]. PNMf does not claim either algorithm is physically causal. They are flexible nonlinear surrogates over the observed descriptor/power space.

Table 6. Exact production learner settings.

Parameter	Extra Trees	Random Forest	Evidence
Estimator class	<code>ExtraTreesRegressor</code>	<code>RandomForestRegressor</code>	<code>pnmf/models.py:90–99</code> [E3]
<code>n_estimators</code>	500	200	[E3, E11]
<code>min_samples_leaf</code>	1	2	[E3, E11]
<code>max_depth</code>	24	None	[E3, E11]
<code>max_features</code>	0.5	1.0	[E5, E11]
<code>n_jobs</code>	-1	-1	[E5, E11]
<code>random_state</code>	Run seed	Run seed	[E5, E11]
Post-prediction constraint	Isotonic non-increase with distance	Isotonic non-increase with distance	<code>pnmf/models.py:65–73, 180–190</code> [E3]

The validation run did not tune these hyperparameters. The run manifest serialises every effective scikit-learn constructor parameter, including defaults [E5]. This prevents selective reporting of only the explicitly written settings.

5.2 Distance-shape constraint

After prediction, PNMf checks each power row. If any level increases by more than `1e-9` between adjacent standard distances, it fits an isotonic regression against `log10(distance)` with `increasing=False` [E3, 9]. The result is a least-squares projection onto a non-increasing

sequence at the ten tabulated distances. Rows already monotone are unchanged.

This is a physically motivated shape constraint, not extra accuracy evidence. It can remove an implausible local increase while also hiding a model's unconstrained error. Validation scores in the current workflow apply the production projection, so they represent the delivered model, not raw tree outputs [E5].

5.3 Tree dispersion

For a query matrix $\mathbf{X}$, PNMf computes

$$\mathbf{s}_j(\mathbf{x}) = \operatorname{std}_{t=1,\dots,T} \left[\hat{y}_{t,j}(\mathbf{x}) \right].$$

Here, $\mathbf{x}$ is one query vector, t indexes the T fitted trees, j indexes one of the ten distance outputs, and $\hat{y}_{t,j}$ is tree t 's prediction for output j . The implementation stacks every tree prediction and applies `std(axis=0)` [E3]. This quantity measures ensemble disagreement under the fitted tree construction. It is useful as a heuristic flag for sparse or extrapolative queries, but it is not a calibrated standard error, prediction interval, confidence probability, or physics uncertainty. No coverage calibration is performed in the current evidence.

5.4 Baseline NPD lookup

For a requested power P and distance d , the `NPDTable` first interpolates or extrapolates each bracketing power row in `log10(d)`, then interpolates or extrapolates linearly in P [E2]. In compact form,

$$L(P, d) = L(P_i, d) + \frac{P - P_i}{P_{i+1} - P_i} [L(P_{i+1}, d) - L(P_i, d)].$$

Here, P_i and P_{i+1} bracket requested power P , and each $L(P_i, d)$ is obtained linearly between ordinates that bracket $\log_{10} d$. This matches the ECAC Doc 29 Section 4.2 baseline convention [1, equations (4-3) to (4-5); E2, E10].

The alignment is limited. Doc 29 specifies resetting a negative power extrapolation value to 1 lbf per engine and recommends a 30 m lower distance bound in the rapidly rising near field [1, printed pp. 28-29]. `NPDTable.level` performs neither safeguard [E2, E18]. It also does not supply the speed/duration, installation/directivity, lateral attenuation, finite-segment, blockage, ground, terrain, or non-uniform-atmosphere terms required elsewhere in a complete Doc 29 chain [1, Sections 4.3-4.5]. The correct claim is "uses the Doc 29 baseline NPD interpolation convention," not "implements Doc 29."

6. Independent component-physics methodology

6.1 Scientific rationale and model identity

The physics route decomposes sound into labelled jet, fan, optional core, wing trailing-edge, slat, flap, and landing-gear contributions. Component decomposition is defensible for sensitivity analysis because source strength depends on different physical quantities and operating states. However, historical source methods have limited data domains and nontrivial correction machinery. Heidmann's fan method uses mass flow, stage temperature rise, Mach/directivity functions, discrete tones, combination tones, and tabulated corrections [5]. Stone separates several flight effects in jet mixing noise [6]. Yamamoto et al. report material discrepancies for historical Fink and Munson airframe methods, especially for landing gear [7].

PNMF therefore uses deliberately qualified language: its equations are reduced Stone-, Heidmann-, and Fink-informed relationships. They are not literal reproductions. The detailed fan path uses simplified directivity, a haystack spectrum, fixed harmonic placements, and a reduced buzz-saw condition rather than all Heidmann functions [E7]. The jet path uses a velocity-to-the-eighth-type source strength, reduced directionality, and calibrated spectral placement rather than Stone's complete flight-effect functions [E7]. Airframe formulas retain conceptual geometry, speed, spectrum, and directivity dependencies but are not validated as generic Fink coefficients for contemporary aircraft [E7, E18]. The precise original Fink report number remains unresolved in local material; no exact Fink provenance or coefficient identity is asserted.

6.2 One-third-octave source spectra

The model evaluates 24 one-third-octave bands [E7]. Reduced source-level relationships include:

- jet source level proportional to `80 log10(V_jet/c0) + 10 log10(A_jet)` plus density, directivity, and calibrated terms (`pnmf/physics.py:493-525`);
- detailed fan level proportional to `10 log10(mdot) + 20 log10(delta_T)` plus directivity and spectral terms (`pnmf/physics.py:591-593`);
- fallback fan context equivalent to `10 log10(mdot) + 40 log10(M_tip)` because Euler work gives `delta_T` proportional to tip speed squared (`pnmf/physics.py:567-568, 629-633`);
- wing/slat terms with approximately `50 log10(V/c0)` and geometry (`pnmf/physics.py:646-680`);
- flap terms with approximately `50 log10(V/c0) + 10 log10(S_flap sin^2(delta_flap))` (`pnmf/physics.py:691-700`); and
- gear terms with approximately `60 log10(V/c0)` and wheel/strut geometry (`pnmf/physics.py:705-725`) [E7].

These are implementation formulas, not universal empirical laws. Every additive anchor and spectral placement is calibration-dependent. Component levels are summed energetically:

$$L_{\text{sum}} = 10 \log_{10} \left(\sum_k 10^{L_k/10} \right),$$

where L_k is the level of contributing source or band k , as implemented for spectra, bands, components, and engine count in `pnmf/physics.py:730-735, 1021-1024, 1148-1172` [E7]. This operation is appropriate for incoherent energy addition; it does not represent phase-coherent interference.

6.3 Propagation, A-weighting, and event metrics

At emission angle θ , the reference path maps closest distance d to range $r=d/\sin(\theta)$ and along-track location $x=-d/\tan(\theta)$. For every one-third-octave band, PNMf applies spherical spreading, molecular absorption, and A-weighting:

$$L_{A,f}(r) = L_f(1, \mathbf{m}) - 20 \log_{10}(r)$$

$$\alpha_f r + A_f$$

where r is source-observer range in metres, $L_f(1, \mathbf{m})$ is band f 's source level at 1 m, α_f is the atmosphere-dependent absorption coefficient, and A_f is the implementation's A-weighting value (pnmf/physics.py:30–83, 1007–1017) [E7]. Band energies are summed to a component time history; engine-source levels receive $10 \log_{10}(n_{\text{eng}})$; component energies are then summed to total $L_A(t)$ (pnmf/physics.py:1148–1172) [E7].

The reported event metrics are:

$$L_{A,\max} = \max_t L_A(t),$$

$$L_{AE} = 10 \log_{10} \left[\frac{1}{t_0} \int 10^{L_A(t)/10} dt \right], \quad t_0 = 1, \text{ s.}$$

Here, $L_A(t)$ is the total A-weighted event history, $L_{A,\max}$ is reported as LAm_{ax}, and L_{AE} is reported as SEL (pnmf/physics.py:1092–1097) [E7].

The code uses trapezoidal integration. The 1 s reference is dimensionally implicit in the numerical definition because time is represented in seconds [E7]. Tests verify inverse-distance propagation, atmospheric behavior, the distinct distance decay of SEL and LAm_{ax}, energetic conservation, and diagnostic retention [E13].

6.4 Explicit exclusions

The EventDiagnostics contract reports these excluded effects: installation/shielding, nacelle treatment, ground reflection, lateral attenuation, terrain, and non-uniform atmosphere [E7]. The model does not calculate EPNL or PNLTm because the required perceived-noise and tone-correction machinery is absent; requests raise ValueError [E7, E13]. It also does not currently demonstrate complete core, turbine, shock, combustion, shielding, scattering, ground, or trajectory mechanisms for arbitrary concepts.

Consequently the physics route is best interpreted as an idealised source and flyover-event screen. A downstream airport model would need to implement, configure, and verify the missing corrections and scenario logic independently.

6.5 A320-270N frozen calibration lifecycle

Calibration fits five quantities: four component anchor constants (C_{jet}, C_{fan}, C_{wingflap}, C_{gear}) and one shared spectral-placement scale. The residual vector pools A320-270N SEL and LAm_{ax} truth for approach and departure across available powers and distances. A declared spectral-scale grid is followed by bounded least-squares refinement [E7, E20]. The release artifact records the anchor aircraft/NPD/preset, optimizer settings, source hashes, code revision, command, timestamp, and in-sample metrics. The current artifact covers 180 cells with RMSE 6.20 dB, MAE 5.33 dB, bias +0.43 dB, and p90 absolute error 9.81 dB [E19, E20].

"Frozen" means that the serialized parameters are loaded once per predictor instance and then reused. NoisePredictor._calibrated_physics() validates the schema, registry version, and source hashes before applying them; it never calls calibrate() and never falls back to A320-211 [E4, E20]. The explicit build-physics-calibration command writes a caller-selected candidate path. Reproduction compares the candidate structurally and numerically with the release artifact. The calibration fit is in-sample and is not external validation.

7. Output contract and engineering interpretation

7.1 Contracted output surfaces

PNMF has several related output surfaces. They must not be conflated: the in-memory prediction object, strict ANP-layout export, QA/storage result, physics diagnostics, comparison scalar, Streamlit calculation trace, and validation run manifest have different schemas and persistence guarantees.

Table 7. Contracted output surfaces and availability.

Output	Exact representation	Surface and persistence	Evidence
Learned NPD tables	Dictionary keyed by (metric, op_mode); each NPDT _{able} contains ascending power vector $\mathbf{P}$, $(\mathbf{P}, 10)$ level matrix $\mathbf{L}$, ten standard distances, metric, mode, identifier, and power-parameter text	Public in-memory NoisePrediction.tables; no automatic file write	pnmf/core.py:280–327; pnmf/api.py:68–75, 187–235; tests/test_api_power_settings.py:34–59, 62–83 [E2, E4]
Physics NPD tables	Same ten-distance NPDT _{able} shape for SEL or LAm _{ax} only	Public NoisePrediction.physics_table(...) convenience method; also retained in Streamlit physics_result; no automatic standalone export	pnmf/api.py:136–142; pnmf/physics.py:1210–1224; tests/test_physics.py:84–96; tests/test_ui.py:97–110 [E4, E7, E13]

Output	Exact representation	Surface and persistence	Evidence
ANP-style CSV	NPD_ID, Noise Metric, Op Mode, Power Setting, then the ten L_*ft columns rounded to 0.1 dB	Public <code>NoisePrediction.to_anp_csv(path_or_buffer)</code> export and Streamlit download; it does not include uncertainty, QA, or physics diagnostics	<code>pnmf/api.py:77-91</code> ; <code>pnmf/ui.py:1120-1124</code>
Tree-dispersion heuristic	Dictionary keyed by <code>(metric, op_mode)</code> with <code>(P,10)</code> standard-deviation arrays aligned to table rows	Public in-memory <code>NoisePrediction.uncertainty</code> ; not part of strict ANP CSV. A separate companion CSV exists only in the <code>demo</code> CLI workflow, not as the canonical export contract	<code>pnmf/api.py:68-75</code> , <code>173-235</code> ; <code>pnmf/models.py:189-198</code> ; <code>pnmf/cli.py:547-570</code> ; <code>tests/test_api_power_settings.py:54-59</code> [E3, E4]
QA result and storage status	<code>(status, reasons)</code> with status <code>ok</code> , <code>caution</code> , or <code>rejected</code> ; stored rows carry <code>qa_status</code> and <code>qa_notes</code>	Internal/public function <code>qa_check</code> ; used by CLI/UI and <code>PredictionStore</code> . It is not a field on a newly returned <code>NoisePrediction</code> until a caller evaluates the table	<code>pnmf/anp.py:576-618</code> , <code>625-646</code> , <code>660-725</code> ; <code>tests/test_learning.py:96-164</code> [E1]
Physics event diagnostics	<code>EventDiagnostics</code> : LAmix, SEL, time vector, total and component time histories, component metrics, source/input status, exclusions, and uncertainty note	Public <code>NoisePrediction.physics_diagnostics(...)</code> and <code>PhysicsNPDModel.single_event_diagnostics(...)</code> ; displayed and retained in Streamlit session state for the active session	<code>pnmf/api.py:120-134</code> ; <code>pnmf/physics.py:984-1001</code> , <code>1118-1184</code> ; <code>tests/test_physics.py:236-255</code> ; <code>tests/test_ui.py:85-110</code> [E4, E7, E13]
Learned-versus-physics comparison	Dictionary <code>{metric: mean absolute delta in dB}</code> over all shared table cells, for SEL and LAmix	Public <code>NoisePrediction.crosscheck_physics(...)</code> ; scalar per metric, not a cellwise residual array or truth error. UI/CLI may display or pass it to QA	<code>pnmf/api.py:93-118</code> ; <code>pnmf/cli.py:168-184</code> ; <code>pnmf/ui.py:977-983</code> [E4]
Calculation trace	Callback events <code>combo_start</code> , <code>combo_done</code> , <code>prediction_done</code> ; Streamlit translates them into elapsed-time entries and stores them in session state	Callback is public but ephemeral; UI trace is session-local, not a signed run artifact	<code>pnmf/api.py:187-235</code> ; <code>pnmf/ui.py:60-92</code> , <code>670-725</code> ; <code>tests/test_api_power_settings.py:62-77</code> ; <code>tests/test_ui.py:45-54</code> [E4]
Validation provenance	JSON run manifest plus samples, splits, predictions, summaries, fit records, source manifest, Git state, software versions, and per-artifact hashes	Persisted CLI validation artifact; distinct from an ordinary <code>NoisePrediction</code> and from the UI calculation trace	<code>pnmf/validation.py:1021-1161</code> ; <code>tests/test_validation.py:99-135</code> [E5, E11, E16]

7.2 Ten-distance NPD and CSV semantics

An NPD table is a matrix whose rows are per-engine power settings and whose columns are the ten standard slant distances. It carries metric, operation mode, distance vector, curve identifier, and power-parameter text [E2]. Learned prediction returns one such table for every fitted/requested metric and mode. Physics returns the same shape only for SEL and LAmix. An NPD table is not a trajectory, contour, receptor time series, certification test, or population-average exposure result by itself.

`NoisePrediction.to_anp_csv` is the canonical standalone export surface. It writes strict ANP-style columns and rounds levels to 0.1 dB (`pnmf/api.py:77-91`). The export is a representation of generated results, not a truth-table insert and not an EASA-issued ANP record. Tree dispersion, QA reasons, physics comparison values, calculation logs, and input provenance are intentionally absent from that strict CSV. Consumers needing those fields must retain separate companion evidence rather than silently extending the ANP layout. The `demo` command's dispersion companion file is an example workflow, not a guaranteed output of every prediction (`pnmf/cli.py:547-570`).

The learned output represents the conditional pattern learned from the available ANP population under the exact feature transformation and fitted learner. The physics output represents the idealised 160 kt event under the physical inputs, fallbacks, calibrated anchors, and exclusions. Equal table shape does not make the semantics identical.

7.3 Metric interpretation

Table 8. Output metrics.

Metric	Learned route	Physics route	Interpretation boundary
SEL	Yes	Yes	A-weighted sound exposure level for a single-event representation
LAmix	Yes	Yes	Maximum A-weighted level in the event representation
EPNL	Yes	No	Learned ANP target only; physics lacks perceived-noise/tone machinery
PNLTM	Yes	No	Learned ANP target only; physics lacks the required tone-corrected machinery

Approach and departure are separate tasks. Power settings are per engine. Learned distance is stored in feet, while physics converts feet to metres at its boundary and works in SI internally [E2, E7].

7.4 Tree dispersion and QA semantics

Tree standard deviation must be labelled "cross-tree dispersion" or "ensemble disagreement." It is aligned to the power rows and distance columns of its associated learned table, as tested after power sorting in `tests/test_api_power_settings.py:54-59`. It is not a calibrated interval. The strict ANP-layout CSV omits it; the in-memory dictionary is the authoritative ordinary-prediction surface.

`qa_check` rejects a table for non-finite levels, levels outside its configured operational bounds, an increase with distance, or invalid/duplicate power rows (`pnmf/anp.py:576–602`). A rejected table is not written by `PredictionStore`. If hard gates pass, mean tree dispersion above the configured caution value or mean learned-versus-physics disagreement above its configured caution value produces `caution`; the table may be stored with its reasons (`pnmf/anp.py:604–618`, `660–725`). `ok` means none of these configured gates or cautions fired. It does not mean scientifically validated, certified, or accurate to a prescribed predictive threshold. The QA values are operational storage controls and are not adopted as validation pass/fail criteria in this report [E1].

7.5 Physics diagnostics

`EventDiagnostics` is the inspectability contract for one physics event. It retains the total A-weighted time history, source-specific histories, component and roll-up SEL/LAmax values, source and input status, excluded effects, and an uncertainty limitation note (`pnmf/physics.py:984–1001`, `1118–1184`). The public convenience method is available only on a `NoisePrediction` associated with its predictor because it needs the predictor's loaded calibration-artifact physics instance (`pnmf/api.py:120–142`). The direct `PhysicsNPDModel` method is also callable. Streamlit stores the active diagnostic object and a calculation trace in session state; these are UI/session artifacts, not a persistent run-manifest format (`tests/test_ui.py:85–110`).

7.6 Comparison protocol and delta meaning

Only matched metrics, modes, power settings, and distances should be compared. A learned-versus-physics difference is a diagnostic residual, not a direct error unless one side has independent truth status. The physics route must not be automatically rescaled to match the learned route, and the learned route must not use physics output as a target or feature. Where either input status is estimated or unavailable, the comparison must retain that context.

`NoisePrediction.crosscheck_physics` predicts physics tables on the learned table's powers, computes absolute cell differences, concatenates approach and departure cells for each shared metric, and returns their arithmetic mean (`pnmf/api.py:93–118`). Thus `crosscheck["SEL"]` and `crosscheck["LAmax"]` are metric-level mean absolute deltas, not signed bias, not distance-resolved differences, not uncertainty, and not a validation score against truth. Physics component/model-form uncertainty remains qualitative. Combining comparison deltas and tree dispersion as though they were commensurate standard deviations is unsupportable.

7.7 Calculation and run provenance

Ordinary API prediction can emit structured progress events containing table index/count, metric, mode, canonical power rows, output shape, and whether dispersion was produced (`pnmf/api.py:201–235`). The Streamlit calculation log adds elapsed time and human-readable stage text, persists it only in `st.session_state`, and is tested for learned completion (`pnmf_ui.py:60–92`, `686–725`; `tests/test_ui.py:45–54`). This is operational traceability, not durable scientific provenance.

Each fitted surrogate also retains an internal `training_provenance` mapping of source-dataset row counts by metric/mode and refuses a production fit with no v6.3 samples (`pnmf/models.py:123–140`). The public prediction metadata additionally records learner, `training_scope`, registry version, selected IDs, support counts, and source hashes. When the physics route is invoked, the metadata records the calibration artifact schema, anchor, parameters, metrics, code revision, and hashes. These records belong to the prediction object and validation bundle; they are not silently appended to the strict ANP CSV.

Durable validation provenance is provided by the validation artifact bundles: source-manifest copy, deterministic samples and splits, predictions, balanced summaries, fit records, exact effective model parameters, Git commit/dirty state, runtime versions, input hashes, and per-generated-artifact hashes [E5, E6, E11, E12]. An ordinary exported NPD CSV does not automatically carry that bundle. A defensible handoff must associate the CSV with aircraft inputs, model name, code/worktree identity, datastore hash, source-manifest hash, power contract, and any QA, comparison, or diagnostic companions required by its use.

8. Software verification

8.1 Verification strategy

Verification asks whether the code implements its stated contracts. It is distinct from validation, which asks whether the scientific output adequately represents the real world for an intended use. PNMF verification combines unit tests, integration tests, data-contract tests, deterministic artifact generation, manifest hashing, and targeted command execution.

Table 9. Verified software and data contracts.

Contract	Verification evidence	Verdict
Legacy plus v6.3 counts and provenance	Exact count assertions and mandatory-source errors in <code>tests/test_migration.py:16–47</code>	Verified
Caller-CWD-independent manifest command	Subprocess from a temporary directory in <code>tests/test_migration.py:62–67</code>	Verified
Only ET/RF supported	Constructor and API rejection tests in <code>tests/test_migration.py:50–59</code> and <code>tests/test_validation.py:79–96</code>	Verified
Power-unit transformations	<code>tests/test_pnmf.py:99–115</code>	Verified
Linear-power/log-distance lookup	<code>tests/test_pnmf.py:69–98</code>	Verified
Group and 7773ER leakage control	<code>tests/test_validation.py:12–44</code>	Verified
Metric formulas	<code>tests/test_validation.py:47–76</code> ; <code>tests/test_jet_reference_validation.py:94–128</code>	Verified
Target-blind frozen jet selection and split	<code>tests/test_jet_reference_validation.py:24–91</code>	Verified
Physics SEL/LAmax-only scope	<code>tests/test_physics.py:84–96</code>	Verified
Typed input status and component gating	<code>tests/test_physics.py:123–215</code> , <code>259–284</code>	Verified
Component and total energetic sums	<code>tests/test_physics.py:218–256</code>	Verified
Versioned validation artifact set	<code>tests/test_validation.py:99–135</code> ; <code>tests/test_jet_reference_validation.py:131–186</code>	Verified

Contract	Verification evidence	Verdict
Exact 14-entry verified registry and 11/3 status split	<code>tests/test_verified_anp.py</code> , <code>tests/test_training_scopes.py</code> , and <code>pnmf/verified_anp.py</code>	Verified
Fail-closed verified provenance, eight-task completeness, and source hashes	<code>tests/test_verified_anp.py</code> and <code>pnmf/verified_anp.py</code>	Verified
Scope-qualified ET/RF metadata, CLI selector, UI registry display, and storage identity	<code>tests/test_training_scopes.py</code> , <code>tests/test_ui.py</code> , <code>pnmf/models.py</code> , <code>pnmf_cli.py</code> , <code>pnmf_ui.py</code>	Verified
Eleven physics presets with field-level provenance	<code>tests/test_physics_presets.py</code> and <code>pnmf/physics_presets.py</code>	Verified
Serialized A320-270N calibration schema, hash rejection, and no runtime refit	<code>tests/test_physics_calibration.py</code> , <code>pnmf/physics_calibration.py</code>	Verified

8.2 Assurance execution results

The current full test suite was rerun on 2026-08-13 with `projects/pnmf/pnmf.ps1 test`: 125 tests passed and 1 test was skipped in 70.87 s [E14]. The skip was `tests/test_learning.py:74`, caused by ANP CSV files not being staged in SQLite-only mode. The historical bounded attempt and its earlier nontermination are retained in E14 for auditability, but are not used as the current suite result.

The verified-scope and calibration-focused lanes completed with 26 passed and 1 skipped for learned/API/UI behavior, 20 passed for core/registry behavior, and 48 passed for physics, presets, calibration, and existing physics contracts. The fresh scope validation command completed with 16,800 grouped, 4,640 frozen-reference, 16,000 temporal, and 40 physics validation rows [E19, E20]. These numeric outcomes apply to the recorded commands and environment only.

Table 10. Test execution status.

Lane	Result	Assurance interpretation
Full suite current rerun	125 passed, 1 skipped in 70.87 s	Terminating software test result for the reviewed worktree [E14]
Historical bounded/full-suite attempt	Earlier nontermination at one component-physics Streamlit test	Retained as historical execution evidence only [E14]
Bounded suite excluding 3 physics UI tests	87 passed, 1 skipped, 3 deselected in 49.36 s	Broad non-UI and remaining UI evidence; three tests intentionally absent [E14]
Focused physics/preset/CI	29 passed in 1.94 s	Targeted physics contract evidence [E14]
Validation/ZIP	8 passed in 26.58 s	Validation artifact and packaging contract evidence [E14]
Verified scope/API/UI lane	26 passed, 1 skipped	Default verified scope, merged comparator, metadata, selector, and prediction identity [E20]
Registry/core lane	20 passed	Exact registry contents, fail-closed resolver, and unchanged core behavior [E20]
Physics/preset/calibration lane	48 passed	Eleven presets, provenance, SEL/Lamax scope, artifact loading, hash mismatch rejection, and no runtime refit [E20]

8.3 Reproducibility of generated evidence

Fresh validation outputs were compared with dated outputs by key and row count. Learned summary keys and all 1,776 rows matched; the maximum numeric difference was approximately $4.09\text{e-}14$. Jet summary keys and all 144 rows matched; the maximum numeric difference was approximately $5.06\text{e-}14$ [E15]. These results demonstrate numerical reproduction within floating-point precision. They are not byte identity.

Fixed random seeds constrain stochastic construction, but multi-threaded scikit-learn execution and floating-point reduction order can change last bits. Timestamps, durations, dirty-state hashes, absolute paths, and run-specific artifact hashes also vary. The run manifests correctly describe hashes as integrity records for a specific execution, not a guarantee that independent executions produce identical bytes [E5, E6].

9. Learned empirical validation

9.1 Metrics

For every predicted cell, error is defined as

$$e_n = \hat{y}_n - y_n,$$

The pooled metrics are

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N e_n^2}, \text{MAE} = \frac{1}{N} \sum_{n=1}^N |e_n|, \text{bias} = \frac{1}{N} \sum_{n=1}^N e_n, p_{90} = Q_{0.90}(|e_1|, \dots, |e_N|).$$

Here, $\hat{y}_n$ is the predicted value for scored cell n , y_n is the truth value for scored cell n , N is the number of pooled cells, and $Q_{0.90}$ is the empirical 90th percentile [E5, E11].

Cell pooling weights every power-distance cell equally. Curve-balanced and aircraft-group-balanced forms first calculate within-unit statistics and then weight units equally. Because cells within an NPD curve are correlated, cell count must not be interpreted as the independent sample count.

No predictive pass/fail threshold is imposed in this report. The values are descriptive evidence to be judged against a declared application and uncertainty tolerance.

9.2 Three-fold aircraft-grouped validation

The split is built from connected components of the bipartite `ACFT_ID` to `NPD_ID` graph. This keeps a curve in one fold, keeps aircraft sharing a curve together, and keeps repeated identities across releases together. The combined inventory has 121 aircraft groups and 122 curves. For

folds 0/1/2, the test partitions contain 40/41/40 groups and 41/41/40 curves; the train complements contain 81/80/81 groups and 81/81/82 curves [E5, [run_manifest.json.fit_runs](#)].

Power-row counts are mode-specific, not a single fold count. For every approach metric, folds 0/1/2 use 225/232/227 training rows and 117/110/115 test rows. For every departure metric, they use 300/299/315 training rows and 157/158/142 test rows [E5, [run_manifest.json.fit_runs](#)]. ET and RF use the same split inventory. The protocol controls exact identifier leakage but does not hold out curated aircraft, engine, or architecture families.

Table 11. Three-fold aircraft-grouped cell-pooled results. Source: fresh [summary.csv](#) [E5].

Model	Metric	Approach RMSE / MAE / bias / p90 (dB)	Departure RMSE / MAE / bias / p90 (dB)
ET	SEL	3.878 / 2.546 / 0.085 / 5.676	4.548 / 3.370 / 0.556 / 6.998
ET	LAmx	4.228 / 2.811 / 0.274 / 6.338	4.661 / 3.533 / 0.579 / 7.356
ET	EPNL	4.550 / 3.086 / 0.174 / 7.019	4.876 / 3.601 / 0.570 / 7.797
ET	PNLTM	4.752 / 3.238 / 0.093 / 7.593	5.045 / 3.810 / 0.408 / 8.183
RF	SEL	3.955 / 2.640 / 0.097 / 5.517	4.677 / 3.481 / 0.275 / 7.069
RF	LAmx	4.262 / 2.894 / 0.206 / 6.529	4.778 / 3.687 / 0.303 / 7.532
RF	EPNL	4.631 / 3.187 / 0.112 / 7.327	5.070 / 3.825 / 0.298 / 8.006
RF	PNLTM	4.896 / 3.372 / 0.139 / 7.730	5.257 / 3.987 / 0.153 / 8.129

The smallest grouped RMSE in Table 11 is 3.878 dB for ET SEL/approach; the largest is 5.257 dB for RF PNLTM/departure [E5]. ET is lower in RMSE for every listed task, but this validation was not designed as a formal model-selection significance test. Both learners remain supported.

Support slicing classifies an exact engine-type/count test slice as impossible when the training fold has zero groups, sparse with one or two, and feasible with at least three. The internal workflow produced 136 feasible, 24 sparse, and 16 impossible exact-cell slices [E5]. A prediction can still be emitted for an impossible exact cell by borrowing across other types or counts, but that is extrapolation rather than evidence of supported generalisation.

9.3 Verified-scope grouped validation and scope comparison

The fresh verified protocol filters the supervised surface to the eleven trainable v6.3 registry IDs and assigns those groups to three deterministic folds with seed 20260724. Each held-out fold is scored at its per-aircraft macro grain as well as by pooled cells. The merged comparator is evaluated on the identical held-out folds after removing held-out NPD IDs and descriptor-identical aliases from its training rows. This controls the comparison boundary without treating the 10 distance columns or multiple power rows as independent aircraft [E19].

Table 13. Fresh verified grouped results at macro-aircraft grain. Values are RMSE / MAE / bias / p90 absolute error in dB; support counts are test power rows and aircraft groups.

Scope	Learner	SEL/A	SEL/D	LAmx/A	LAmx/D	Support
Verified	ET	2.307 / 1.797 / -0.043 / 2.542	3.404 / 2.618 / -0.672 / 4.226	2.029 / 1.554 / -0.031 / 2.211	3.616 / 2.725 / -0.349 / 4.726	46/59 rows; 11 groups
Verified	RF	2.270 / 1.873 / -0.061 / 2.662	3.639 / 2.845 / -0.664 / 4.852	2.072 / 1.679 / -0.065 / 2.523	3.807 / 2.976 / -0.526 / 5.181	46/59 rows; 11 groups
Merged comparator	ET	2.301 / 1.426 / +0.351 / 2.054	3.803 / 2.582 / -0.222 / 3.967	2.092 / 1.184 / +0.459 / 1.742	3.833 / 2.629 / -0.173 / 4.455	same held-out groups
Merged comparator	RF	2.363 / 1.730 / +0.774 / 2.483	4.037 / 3.340 / +0.496 / 5.128	2.227 / 1.437 / +0.904 / 2.138	4.207 / 3.322 / +0.350 / 5.546	same held-out groups

The full eight-task table, scope deltas, support counts, fit records, and cell-level predictions are in [summary.csv](#), [verified_vs_merged_delta.csv](#), [verified_group_runs.csv](#), and [predictions.csv](#) under E19. No predictive pass threshold is inferred from these residuals. The result supports only retrospective bounded validity for the eleven represented conventional families.

9.4 Release-ordered temporal holdout

The temporal protocol fits legacy v2.3 targets and evaluates v6.3 targets. The raw version includes exact identity overlap. The purged version removes the v6.3 7773ER test curve because that identity occurs in legacy training, while leaving training unchanged. The raw test has 11 curves; the purged test has 10 [E5].

Table 12. Purged legacy-to-v6.3 temporal holdout RMSE (dB). Source: fresh [summary.csv](#) [E5].

Model	SEL A	SEL D	LAmx A	LAmx D	EPNL A	EPNL D	PNLTM A	PNLTM D
ET	2.419	4.640	2.217	4.639	2.761	5.290	2.645	5.074
RF	2.467	4.920	2.359	4.950	2.917	5.606	2.522	5.072

Approach errors are lower than departure errors in this small holdout [E5]. The result may reflect task difficulty, source composition, power-grid differences, or aircraft mix; it does not isolate a causal explanation. The holdout is chronological by staged release label, not by physical aircraft-family novelty. The supplement is small and contains related conventional aircraft. Therefore the experiment provides release-transfer evidence but not prospective external validation.

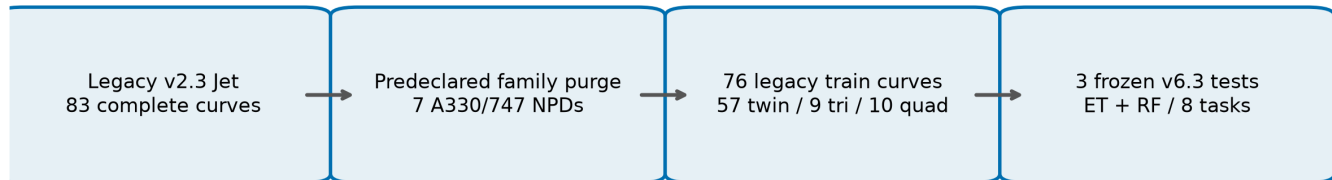
10. Frozen v6.3 jet reference validation

10.1 Target-blind selection and separation

The frozen jet experiment restricts candidates to v6.3 Jet curves with all eight tasks. Within two-, three-, and four-engine categories, it calculates medians and interquartile ranges for [log10\(MTOW_lb\)](#), [log10\(total static thrust_lb\)](#), and noise chapter. Candidate distance is

$$D(\mathbf{x}) = \sqrt{\sum_j \left(\frac{x_j - \tilde{x}_j}{\text{IQR}_j} \right)^2},$$

where x_j is candidate descriptor j , $\tilde{x}_j$ is its category median, and IQR_j is its category interquartile range; a zero-IQR contribution is defined as zero [E6, E12]. Noise targets, predictions, and errors are excluded from selection. The selected references are A330-743L (two engines, distance 0.216692), FAL900EX (three engines, distance 0), and 747400RN (four engines, distance 0) [E6]. The zero distances for the latter two arise because each category has only one candidate; they do not prove category representativeness.



v6.3 targets are absent from training and descriptor selection never uses noise levels; the physics route remains independent.

Jet reference validation architecture

Figure 3. Frozen v6.3 jet validation architecture. Source: fresh 2026-08-13 jet-reference run; the embedded repository image is byte-identical to the fresh artifact [E6].

A predeclared family guard removes seven related legacy curve identifiers: CF680E, TRENT7, JT9DBD, JT9DFL, JT9D7Q, PW4056, and GENX67 [E6]. The final training set contains 76 curves and 2,052 metric/mode power rows: 57 twins, 9 tri-jets, and 10 quad-jets. The test set contains 3 curves and 116 metric/mode power rows, one curve per engine-count category. No v6.3 target row enters training [E6, E12].

The test contains 116 power rows and 1,160 power-distance cells across four metrics and two modes. These are correlated cells from three curves, not 1,160 independent aircraft observations [E6].

10.2 Overall results

Table 13. Frozen jet holdout overall results. Source: fresh jet-reference-current/summary.csv [E6].

Model	Aggregation	RMSE (dB)	MAE (dB)	Bias (dB)	p90 absolute error (dB)	Within +/-3 dB (%)	Within +/-5 dB (%)
ET	Category-balanced	3.931	3.213	0.459	6.010	50.648	73.343
ET	Cell-pooled	3.953	3.239	0.545	6.425	50.000	72.931
RF	Category-balanced	3.868	3.193	0.413	5.785	51.972	74.491
RF	Cell-pooled	3.900	3.231	0.491	6.398	51.121	73.966

Category-balanced RMSE is the square root of the mean of category mean-squared errors, giving the three engine-count categories equal weight [E6, E12]. The percentages within +/-3 dB and +/-5 dB are descriptive threshold-agreement rates over correlated cells. They are not aircraft-level success rates, confidence probabilities, certification margins, or report-imposed acceptance criteria.

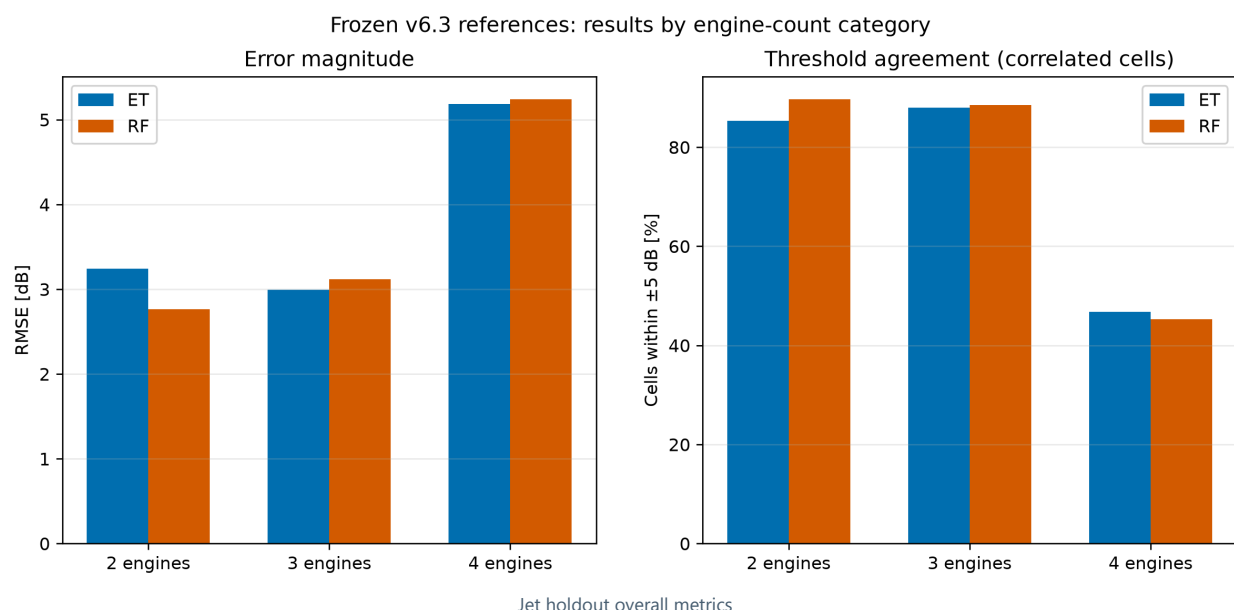


Figure 4. Frozen jet holdout overall metrics. Source: fresh 2026-08-13 jet-reference run; repository image verified byte-identical to the fresh artifact [E6].

10.3 Category behavior

Table 14. Frozen jet holdout cell-pooled results by reference category. Source: fresh summary.csv [E6].

Model	Engine-count category	RMSE (dB)	MAE (dB)	Bias (dB)	p90 absolute error (dB)	Cells
ET	2	3.240	2.440	-2.021	5.972	360
ET	3	2.993	2.232	-1.562	5.386	400
ET	4	5.187	4.966	4.961	6.672	400
RF	2	2.765	2.087	-1.855	5.222	360
RF	3	3.121	2.503	-1.894	5.271	400
RF	4	5.244	4.987	4.987	6.863	400

Both learners underpredict the selected twin and tri-jet on average and overpredict the selected quad-jet on average [E6]. Because the tri- and quad-engine categories each contain one v6.3 candidate, the pattern is a three-reference case study rather than an engine-count population estimate.

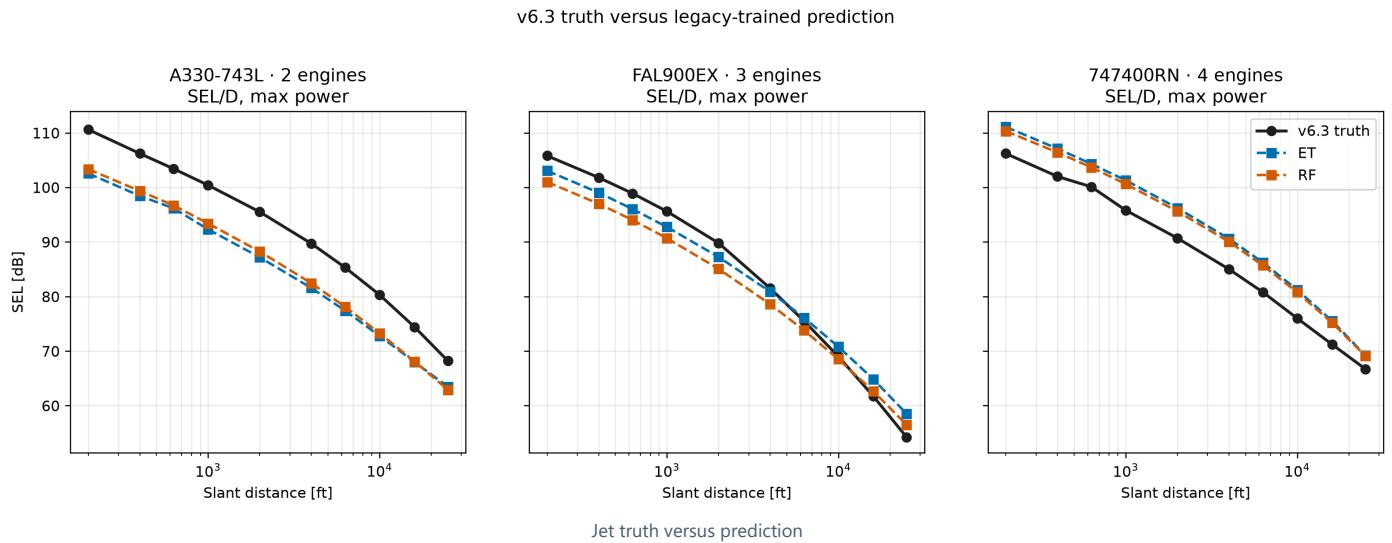


Figure 5. Truth versus prediction for the three frozen v6.3 jet references. Source: fresh 2026-08-13 jet-reference run; repository image verified byte-identical to the fresh artifact [E6].

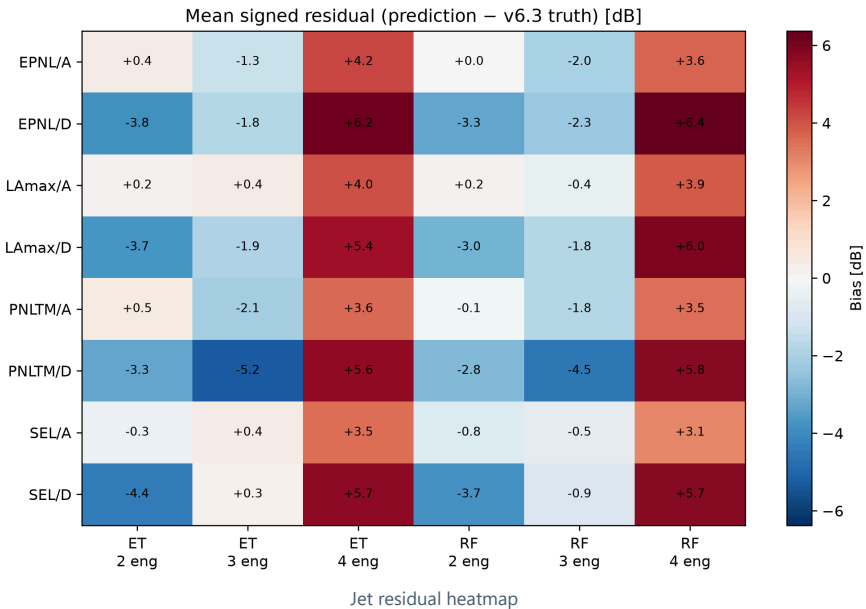


Figure 6. Residual heatmap for the frozen v6.3 jet references. Source: fresh 2026-08-13 jet-reference run; repository image verified byte-identical to the fresh artifact [E6].

The frozen jet experiment is stronger than ordinary random cell splitting because targets are release-separated, selection is target-blind, and obvious A330/747 family analogues are purged. It is still insufficient for a general external-validation claim. Only three test curves exist, selection descriptors omit detailed engine technology and geometry, and conservative name-based purging cannot prove engineering independence.

10.4 Verified models trained on the other eight families

The new verified-scope frozen case uses the same three target-blind jet IDs, but trains a verified model on the other eight verified families and a merged comparator after removing the target groups and descriptor-identical aliases. It reports all eight metric/mode tasks for ET and RF. The machine-readable results are the `frozen_verified_jets` rows in E19; the test contains three aircraft and correlated power-distance cells. This case does not convert the three aircraft into an external population and does not supersede the historical legacy-trained experiment above.

11. Physics evidence

11.1 Calibration and fleet comparison

The fresh physics artifact reports an A320-270N in-sample calibration over 180 cells: RMSE 6.20 dB, MAE 5.33 dB, bias +0.43 dB, and p90 absolute error 9.81 dB [E19, E20]. This fit is not external validation. With the serialized parameters held fixed, the remaining ten trainable verified families are scored for SEL and LAmax in approach and departure. The results are stratified by field provenance and fallback status rather than collapsed into a certification-style pass rate.

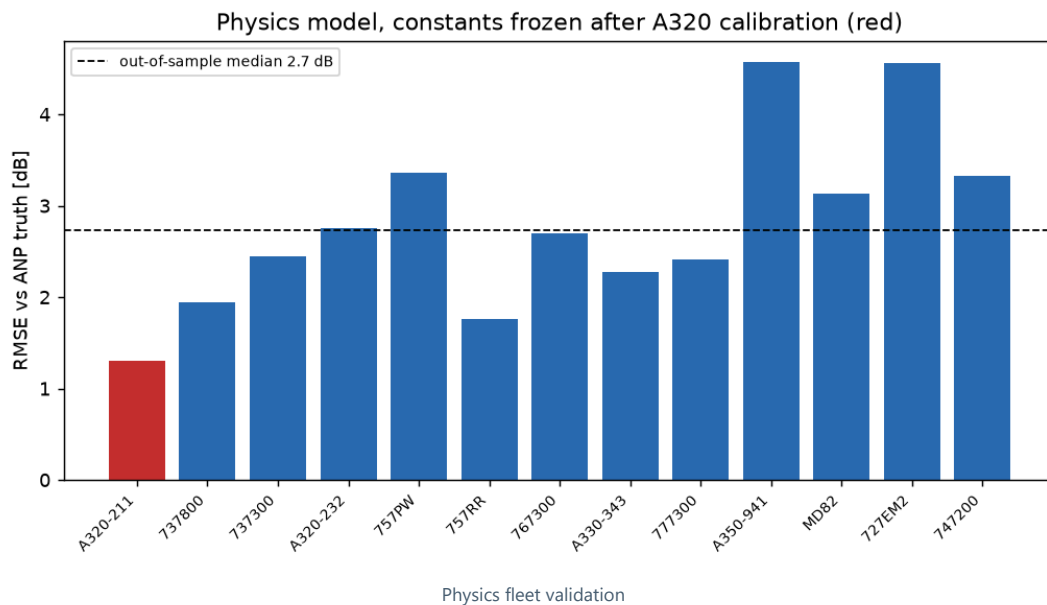


Figure 7. Physics fleet comparison with constants fixed after A320-270N calibration. The numeric authority for the revised run is `physics_validation.csv` [E19]; the historical figure is retained only as a dated comparator [E8].

Table 15. Revised frozen-physics validation summary. Source: `physics_validation.csv` [E19]. Values are RMSE / MAE / bias / p90 absolute error in dB; each row represents one aircraft/metric/mode task.

Aircraft examples	Metric/mode	RMSE / MAE / bias / p90 (dB)	Input provenance and fallback
A321-270N	SEL/A	6.83 / 6.13 / -6.05 / 10.00	8 supplied, 5 estimated, 2 unavailable; fallback
A330-743L	SEL/D	10.58 / 10.09 / +10.09 / 14.30	8 supplied, 5 estimated, 2 unavailable; fallback
A350-1041	LAmax/D	4.14 / 3.28 / +0.79 / 6.94	9 supplied, 4 estimated, 2 unavailable; fallback
747400RN	SEL/D	18.81 / 18.45 / +18.45 / 22.32	8 supplied, 5 estimated, 2 unavailable; fallback
7673ER	LAmax/A	6.08 / 5.46 / -5.45 / 8.64	8 supplied, 5 estimated, 2 unavailable; fallback
7879	SEL/D	6.93 / 6.06 / +5.46 / 10.93	9 supplied, 4 estimated, 2 unavailable; fallback
7773ER	LAmax/D	4.50 / 3.76 / -1.73 / 7.17	9 supplied, 4 estimated, 2 unavailable; fallback
ERJ190-300	SEL/A	6.25 / 6.17 / -6.17 / 7.64	8 supplied, 5 estimated, 2 unavailable; fallback
ERJ190-400	LAmax/A	5.31 / 5.25 / -5.25 / 6.53	8 supplied, 5 estimated, 2 unavailable; fallback
FAL900EX	SEL/D	18.60 / 18.58 / +18.58 / 19.80	6 supplied, 7 estimated, 2 unavailable; fallback

11.2 Evidence interpretation

The ten non-anchor rows in Table 15 are frozen-parameter validation within the verified registry, not independent external validation. The evaluated aircraft are related conventional transports, the model uses public BPR assumptions rather than controlled manufacturer engine decks, several geometric quantities are estimated, and the anchor and comparison truth come from the same broad ANP system [E7, E19]. A320 family proximity also limits independence.

The frozen-physics result supports the narrower claim that one serialized anchor set can be applied reproducibly to the ten remaining trainable families under the implemented reduced model. It does not validate individual component spectra, causal parameter responses, new propulsion architectures, unconventional installations, or operations beyond the reference event. Source-model scientific validation is therefore weaker than the software and data verification.

11.3 Output-isolation caveat

The physics run exposed a reproducibility defect in output isolation. `pnmf_cli.py` resets the current working directory to the project root. As a result, the isolated run refreshed the ignored `projects/pnmf/outputs` files rather than writing directly to the temporary reason-report tree. The fresh artifacts were then copied into `tmp/reason-report/physics-run/outputs` [E14, E17]. This is an artifact-location and

reproducibility defect. It is not evidence for or against physical model validity. Future assurance runs should accept an explicit output directory and record it in a manifest.

12. Claim-argument-evidence assurance matrix

Verdicts use exactly three levels: **Verified**, **Empirically supported within scope**, and **Not established**. "Verified" is reserved primarily for implementation or data-contract claims. It does not imply scientific truth outside the tested contract.

Table 16. Assurance matrix.

Claim	Argument	Evidence	Verdict
C1. Canonical ANP rows retain release and file provenance.	Ingestion adds provenance columns and embeds a manifest.	E1, E9	Verified
C2. v6.3 is required in the default datastore.	Missing/emotv required v6.3 tables raise DataSourceError ; tests exercise the failure.	E1, E9	Verified
C3. Collision handling is deterministic and preserves the two 7773ER aircraft/NPD pairs.	Table-specific business keys and supplement-wins ordering are explicit.	E1, E9	Verified
C4. ANP truth is separated from generated predictions.	Builder writes anp_* ; guarded prediction storage writes only predicted_* .	E1	Verified
C5. The supported learned surface is exactly ET and RF.	Constructors and API reject other names; hyperparameters are asserted.	E3, E4, E9, E11	Verified
C6. Learned samples use the declared 12 inputs and ten targets.	Feature construction and row assembly are explicit; run manifest records feature names.	E2, E3, E5, E16	Verified
C7. Delivered learned curves are non-increasing at standard distances.	Post-prediction isotonic projection enforces the constraint.	E3	Verified
C8. Tree standard deviation is a calibrated interval.	No interval calibration or coverage test exists.	E3, E5	Not established
C9. Baseline lookup follows linear-power/log-distance interpolation.	Code and unit tests implement and check the convention.	E2, E10, [1]	Verified
C10. PNMf is a complete Doc 29 event/contour implementation.	Required safeguards, corrections, and Volume 3 equivalence evidence are absent.	E2, E7, E18, [1], [2]	Not established
C11. Learned and physics fitting are computationally independent.	No learned output enters physics; comparison occurs only after prediction.	E3, E4, E7	Verified
C12. Physics inputs preserve supplied/estimated/unavailable status.	Typed contracts and diagnostic mappings expose status.	E7, E13	Verified
C13. Physics computes one-third-octave propagation, A-weighting, energy sums, SEL, and LAmax as documented.	Direct code inspection and focused tests cover the calculation.	E7, E13	Verified
C14. Physics supports EPNL or PNLTM.	Requests are explicitly rejected; tone machinery is absent.	E7, E13	Not established
C15. Physics is a literal Stone/Heidmann/Fink or ANOPP reproduction.	The implementation contains reduced analytic substitutes and incomplete provenance/functions.	E7, E18, [5]-[7]	Not established
C16. Grouped ET/RF residual performance is characterised for the current corpus.	Deterministic connected-component folds and complete summary artifacts exist.	E5, E11, E16	Empirically supported within scope
C17. Legacy-to-v6.3 transfer is characterised for the current small supplement.	Raw and exact-identity-purged holdouts are reported.	E5, E11, E16	Empirically supported within scope
C18. Frozen jet performance is characterised for the three selected v6.3 references.	Target-blind selection, family purge, source separation, and 1,160 scored cells are recorded.	E6, E12	Empirically supported within scope
C19. Learned models generalise to unseen aircraft families and unconventional architectures.	No curated family labels or prospective unconventional test set exists.	E5, E6	Not established
C20. The verified registry is exactly the reviewed 14-family list with 11 trainable and 3 metadata-only entries.	Versioned registry, pinned hashes, status tests, and unavailable UI rows are recorded.	E19, E20	Verified
C21. Verified grouped ET/RF residual performance is characterised for eleven conventional families.	Fresh three-fold same-fold verified/merged results report macro-aircraft and cell metrics.	E19	Empirically supported within scope
C22. Physics retains frozen-parameter correspondence for the ten non-anchor verified families.	A320-270N is an in-sample anchor; the remaining ten rows record residuals and input/fallback completeness.	E19, E20	Empirically supported within scope
C23. Physics has independent external system and component validation.	ANP anchoring, public BPR assumptions, related aircraft, and absent component measurements prevent the claim.	E7, E19	Not established
C24. The complete software suite terminates with passing tests.	Current rerun records 125 passed and 1 skipped; the skip is an environment/data-staging condition.	E14	Verified
C25. Fresh verified artifacts preserve hashes, counts, model parameters, and run provenance.	E19 includes source, datastore, calibration, split, and code-state records.	E19	Verified
C26. Generated artifacts are guaranteed byte-identical across executions.	Run manifests explicitly limit the guarantee to numerical reproduction and run-specific integrity.	E5, E6, E19	Not established
C27. PNMf output is certification or regulatory compliance evidence.	Intended scope, missing mechanisms, and absent certification process preclude the claim.	E5-E8, E18-E20, [1]-[4]	Not established

13. Limitations and evidence gaps

13.1 Data population and representativeness

ANP is an authoritative modelling resource, but the staged population is not a balanced experimental design for conceptual-aircraft machine learning. Aircraft share NPD curve identifiers, releases overlap in identity, and rare engine-type/count combinations have sparse or impossible exact support [E5]. The data do not include curated platform, engine-family, or architecture labels suitable for true family holdout. Release chronology is not equivalent to engineering novelty.

The ten distance values and multiple power rows within a curve are correlated. Reporting thousands of cells can create false precision if aircraft/curve counts are omitted. All future reports should lead with aircraft groups and curves, then provide cell counts for computational traceability.

13.2 Learned feature sufficiency

The 12 features omit detailed fan pressure ratio, nozzle state, installation, shielding, airframe geometry, flap/slat/gear design, trajectory, atmosphere, and source directivity. Noise chapter is a coarse categorical/numeric proxy. RPM-to-thrust conversion is approximate. Tree ensembles interpolate associations in the observed feature space; they do not guarantee physically correct extrapolation or causal trends.

The distance monotonicity projection constrains shape only. It does not enforce correct power monotonicity, asymptotic spreading, spectral consistency, metric consistency, or a physically valid relationship between approach and departure.

13.3 Uncertainty

Cross-tree dispersion is not calibrated. The validation reports residual distributions but does not deliver conditional prediction intervals. Physics uncertainty is not propagated from input status, component anchors, public BPR assumptions, atmosphere, geometry fallbacks, or model form. Route disagreement is not a calibrated error bar.

13.4 Physics completeness

The current component equations omit or simplify important mechanisms. Installation/shielding/scattering, nacelle treatment, ground reflection, lateral attenuation, terrain, non-uniform atmosphere, realistic segmented trajectories, and tone-corrected EPNL/PNLTM are explicit gaps [E7]. The core path is disabled without a complete state. Public BPR and estimated geometry can dominate apparent sensitivity. Historical Fink method limitations and unresolved report numbering prevent a coefficient-identity claim [7, E18].

The single A320-270N calibration may absorb missing physics into global anchors. Good aggregate residuals can therefore coexist with wrong component levels or wrong causal trends. Component-resolved measured spectra and controlled perturbation cases are needed. The calibration artifact makes the fit reproducible; it does not make the fit externally valid.

13.5 Verification completeness

The historical assurance run recorded three component-physics Streamlit tests that did not complete, but the current full-suite rerun terminates with 125 passed and 1 skipped [E14]. The report also identifies an output-isolation defect in the historical physics CLI run. The current verified-scope validation writes its artifacts below the explicit reason-report directory; the historical defect remains a reproducibility caveat until the CLI itself accepts an output root.

13.6 Standards equivalence and certification

PNMF has not executed the Doc 29 Volume 3 reference cases and does not implement the full Doc 29 event/contour chain [2, E18]. Some ANP entries may draw on certification-test data, but a learned prediction, cross-validation residual, or A320-anchored physics estimate is not a certification measurement [1, Appendix G; 3]. No conclusion about ICAO Annex 16, FAA Part 36, or other certification compliance is established.

14. Prioritized next validation steps

Table 17. Prioritized evidence-development plan.

Priority	Action	Why it is first-order	Completion evidence
1	Curate immutable aircraft/platform/engine-family labels and run leave-family-out ET/RF validation.	Exact-ID grouping does not test unseen families.	Versioned label file, frozen split, per-family residuals, support matrix, and no target leakage
2	Acquire a prospectively frozen external NPD dataset unused in training, feature design, hyperparameter choice, or threshold selection.	This is the clearest route to independent predictive validation.	Data-use protocol, hashes, predeclared metrics, untouched test targets, and signed result manifest
3	Preserve the terminating full-suite result and keep the historical physics UI timeout under regression monitoring.	The current test contract now terminates, while the historical timeout remains an execution-risk signal.	Full-suite result artifact, timeout history, and a repeatable component-physics UI lane
4	Make all validation and physics commands accept an explicit output directory without changing caller CWD.	Current output isolation weakens reproducibility.	Tests from an external CWD, manifest-recorded output root, and no writes to default ignored outputs
5	Obtain external component spectra and prospective system measurements for the serialized A320-270N calibration and at least one non-anchor family.	The artifact removes runtime refitting but does not supply external validation.	Component input ledger, spectral comparisons, directivity cases, and blinded system rollup
6	Obtain provenance-labelled manufacturer or research engine decks and measured component spectra for at least one calibration and one non-calibration aircraft.	Aggregate ANP residuals cannot validate source components.	Component input ledger, spectral comparisons, directivity cases, and blinded system rollup
7	Quantify physics uncertainty through input ranges and model-form alternatives.	Current status labels are qualitative.	Predeclared distributions/ranges, sensitivity indices, and calibrated coverage where data permit
8	Implement and verify missing propagation/installation modules only when primary equations, coefficients, units, domain, and reference cases are available.	Adding global offsets would hide missing mechanisms.	Module-specific tests and reference-case comparisons; no silent recalibration
9	Calibrate learned prediction intervals on held-out groups.	Raw tree spread has no coverage meaning.	Group-held-out conformal or equivalent calibration with empirical coverage and width by support slice
10	Assess Doc 29 Volume 3 reference cases for the portions PNMf intends to claim, clearly separating unsupported chain elements.	Prevents overbroad standards language.	Case-by-case equivalence matrix and reproducible diagnostics [2]

No new global predictive pass/fail threshold should be chosen before the intended application, independent data, loss consequences, and uncertainty method are declared. A threshold selected after viewing the same validation residuals would be optimistically biased.

15. Glossary

Table 18. Glossary.

Term	Meaning in this report
ANP	Aircraft Noise and Performance data resource maintained for aircraft-noise modelling.
A-weighting	Frequency weighting applied to one-third-octave levels before energetic summation to an A-weighted level.
Bias	Arithmetic mean of prediction - truth ; positive values indicate overprediction on average.
BPR	Engine bypass ratio. Several fleet values in the physics exercise are public assumptions, not controlled engine-deck inputs.
Calibration	Estimation of model parameters from reference truth. PNMf physics uses A320-270N SEL/LAmax to fit four anchors and one spectral scale.
Cell	One power-distance target or prediction. Cells within a curve are correlated.
Conceptual screening	Early-stage comparative analysis intended to guide questions and data acquisition, not approve a design.
Corrected net thrust (CNT)	ANP power parameter, represented in lbf per engine or percent form and transformed by the learned input contract.
Curve	A set of NPD power rows for one NPD identifier, metric, and operation mode.
Doc 29	ECAC guidance for computing civil-airport noise contours. PNMf uses only the baseline NPD interpolation convention.
EPNL	Effective Perceived Noise Level. Available only from the learned route.
ET	Extra Trees regressor, one of two supported learned estimators.
Evidence boundary	Explicit statement of what an artifact supports and what remains unproven.
External validation	Evaluation on evidence isolated from training and model-selection feedback. Current PNMf studies do not fully achieve this standard.
Frozen jet reference	Three preselected v6.3 Jet curves used in the release-separated experiment.
Frozen physics calibration	A320-270N anchors stored in a versioned, hash-pinned artifact, loaded once per predictor instance, and reused without runtime refitting.
Grouped validation	Cross-validation that keeps connected aircraft/NPD identities in one fold. It is not the same as family holdout.
LAmax	Maximum A-weighted level during the represented event.
MAE	Mean absolute cell error.
NPD	Noise-power-distance relationship or table.
One-third-octave spectrum	Band representation with three nominal bands per octave; PNMf physics uses 24 bands.
Operation mode	Approach (A) or departure (D), treated as distinct prediction tasks.
PNLTM	Maximum tone-corrected perceived noise level. Available only from the learned route.
Provenance	Recorded source dataset, source file, or supplied/estimated/unavailable status.
RF	Random Forest regressor, one of two supported learned estimators.
RMSE	Square root of mean squared cell error.
SEL	Sound Exposure Level for the represented event.
Source-informed	Motivated by a primary method's variables or structure without claiming literal reproduction.
Temporal holdout	Training on legacy v2.3 targets and testing on v6.3 targets.
Tree dispersion	Standard deviation across individual tree predictions; an ensemble disagreement heuristic, not a calibrated interval.
Truth	Canonical staged ANP target rows, kept separate from generated predicted_* records.
Validation	Evidence that model output is adequate for an intended real-world use. Distinct from software verification.
Verification	Evidence that implementation and data processing satisfy stated contracts.

16. References

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- scikit-learn developers, *ExtraTreesRegressor* API documentation, <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.ExtraTreesRegressor.html>.
- scikit-learn developers, *RandomForestRegressor* and *IsotonicRegression* API documentation, <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html> and <https://scikit-learn.org/stable/modules/generated/sklearn.isotonic.IsotonicRegression.html>.

Appendix A. Reproducibility record

A.1 Frozen run identity

Table A1. Software and evidence environment. Source: current run manifests and assurance execution record [E5, E6, E14].

Item	Value
Git commit	595a25979ed3104ff17f698fbd73488addf6cd8b
Git dirty	True
Dastore SHA-256	9b3ea2f58347ebb348128c49b3238e6a0b28852858fa7e87c89fad51d5e9fe8e
Source-manifest SHA-256	6d14674d0cd46c75cc6cc969b4b0ab22bfc885ad1cc09e3ea2560832b78cf510
Python	3.14.2
numpy	2.5.1
pandas	3.0.5
scikit-learn	1.9.0
scipy	1.18.0
matplotlib	3.11.1
Model seed	20260724
Verified registry	easa-verified-14-family-v1; 11 trainable, 3 verified-metadata-only
Calibration artifact	pnmf/physics_calibration_A320-270N_v1.json; SHA-256 f37e8e05aa61037e7cd40814531d6519642ac24035742315c264941d6e45e618

Because the worktree was dirty, a reproducer needs the tracked commit plus the exact uncommitted diff represented by the run's `git.status_sha256`, not only the commit identifier [E5, E6]. The datastore and source-manifest hashes bind the main model inputs.

A.2 Evidence integrity

Table A2. Selected input and generated-artifact integrity hashes. Source: fresh learned and jet run manifests [E5, E6].

Evidence object	SHA-256	Interpretation
Canonical datastore	9b3ea2f58347ebb348128c49b3238e6a0b28852858fa7e87c89fad51d5e9fe8e	Input database identity for both runs
Historical canonical source-manifest content	316d3362f1b064a54f6be1f026a11c3fe297a0172e4ad95fc44ba4e7e8e0683f	Hash of the ordered source-manifest records used by the dated pre-scope bundles
Learned <code>splits_internal.csv</code>	27b57a268a3e231fc8e5b3b0e2b5f1d32e7fd8734c4a32e88943cecdbe630e07	Grouped-fold assignment artifact
Learned <code>predictions.csv</code>	9490b26ed4e274fa529461c7023ff65be423dad73bba9f6938da2a95fae90710	Fresh learned cell predictions
Learned <code>summary.csv</code>	cc3291d45c046c41b2a1dca034916df5afc3457e8da808eb4ed9c2553184d22e	Fresh learned aggregate metrics
Jet <code>split.csv</code>	7a6b591b4399167cee9cd02797226d1238a4602dd6d9e41c8c0bd8a82b4ebad4	Frozen legacy-train/v6.3-test assignment
Jet <code>predictions.csv</code>	c9d6f0cb3879902d31863b78c71d127036dab4ee141eb8549391fe8f05d97b4e	Fresh frozen-jet cell predictions
Jet <code>summary.csv</code>	ea2197be6fe004d946976d20035641364fcb87543c26047b7367ea5438aed749	Fresh frozen-jet aggregate metrics
Assurance execution record	Historical 7b8e908e485bba674c1200a7eaa1bfff23e7a092784c7dd1557088c1a5fd9c9eb; final full-suite result e141c847f18a19aef910cd64dce4e1430f2e3680703eb0771712ccceca9d116a	Test commands/results, historical timeout, and current terminating suite [E14]
Fresh-versus-dated comparison	a5d71361e29e460af3b30ccb7834a5bfad9b4db3c345abb9413d5f37eecdabc2	Inputs, method, keys, counts, and numeric maxima [E15]

Both bundles contain `source_manifest.csv` with the logical table, source dataset, and source filename entries used by the datastore. Its generated-file SHA-256 is 76ce5b6369d6fed7be01392820d49a126a2198bed0e4b4aa93ee2800d9c854a0 in both manifests [E5, E6]. The run manifests enumerate all remaining generated-artifact hashes; Table A2 selects the split, prediction, and summary artifacts needed to bind the principal reported results without reproducing the entire manifest.

A.3 Exact 2026-08-13 commands and artifact paths

The fresh manifest evidence was produced from the repository root with:

```
projects\pnmf\pnmf.ps1 manifest
```

The two fresh validation commands were run directly from the repository root with the PNMf virtual environment:

```
projects/pnmf/.venv/Scripts/python.exe projects/pnmf/pnmf_cli.py validate-model `
--output-dir tmp/reason-report/model-validation-current `
--report tmp/reason-report/MODEL_TRAINING_REPORT_CURRENT.md `
--folds 3 --seed 20260724

projects/pnmf/.venv/Scripts/python.exe projects/pnmf/pnmf_cli.py validate-jet-reference `
--output-dir tmp/reason-report/jet-reference-current `
--report tmp/reason-report/JET_REFERENCE_VALIDATION_REPORT_CURRENT.md `
--assets-dir tmp/reason-report/jet-reference-assets `
--seed 20260724
```

The validation argument resolver treats these relative output paths as relative to `projects/pnmf`, so the actual bundles are `projects/pnmf/tmp/reason-report/model-validation-current`, `projects/pnmf/tmp/reason-report/jet-reference-current`, and `projects/pnmf/tmp/reason-report/jet-reference-assets`; the reports are the two named Markdown files directly under `projects/pnmf/tmp/reason-report` (`pnmf/validation.py:1170-1201`; `pnmf/jet_reference_validation.py:1011-1042`).

The exact software-verification lanes were:

```
# Current full-suite rerun: 125 passed, 1 skipped in 70.87 s.
powershell.exe -NoProfile -ExecutionPolicy Bypass -File projects/pnmf/pnmf.ps1 test

# Historical bounded attempt retained for auditability; it did not terminate.
# See assurance_execution_record.md for the original command and timeout.

# Bounded suite: 87 passed, 1 skipped, 3 deselected in 49.36 s.
projects/pnmf/.venv/Scripts/python.exe -m pytest -q projects/pnmf/tests -k "not component_physics"

# Focused physics/presets/CI: 29 passed in 1.94 s.
projects/pnmf/.venv/Scripts/python.exe -m pytest -q `
  projects/pnmf/tests/test_physics.py `
  projects/pnmf/tests/test_physics_presets.py `
  projects/pnmf/tests/test_ci_smoke.py

# Validation/ZIP: 8 passed in 26.58 s.
projects/pnmf/.venv/Scripts/python.exe -m pytest -q `
  projects/pnmf/tests/test_validation.py `
  projects/pnmf/tests/test_zip_extractor.py
```

The repo-root equivalent physics command is:

```
projects/pnmf/.venv/Scripts/python.exe projects/pnmf/pnmf_cli.py physics
```

The actual isolation launch used the absolute path to `projects/pnmf/pnmf_cli.py` while the process working directory was temporary. Module startup nevertheless reset the current directory, and `cmd_physics()` hard-codes `outputs/`. The command therefore refreshed `projects/pnmf/outputs`; fresh artifacts were then copied to `projects/pnmf/tmp/reason-report/physics-run/outputs` for this evidence bundle [E14, E17]. No `--output-dir` flag existed for this command, and none is implied here.

The learned evidence bundle must contain `samples.csv`, `splits_internal.csv`, `splits_temporal.csv`, `predictions.csv`, `summary.csv`, `summary.json`, `engine_support_matrix.csv`, `fit_runs.csv`, `source_manifest.csv`, the training report, and `run_manifest.json` [E5, E11]. The jet bundle must contain candidate selection, split, predictions, fit records, summaries, metadata, source manifest, run manifest, and the four figures [E6, E12].

A.4 Artifact interpretation checks

After reproduction:

1. verify datastore and source-manifest SHA-256 values before comparing metrics;
2. verify seed, learner parameters, feature list, targets, and split protocol;
3. compare artifact keys and row counts before numeric values;
4. use a floating-point tolerance rather than demanding byte identity for model results;
5. confirm no v6.3 target row enters frozen jet training;
6. confirm physics fleet numbers against `physics_fleet_validation.csv`, not chart pixels;
7. record Git dirty state and status hash; and
8. record all excluded, skipped, or nonterminating tests, including historical attempts.

The fresh-versus-dated comparison on 2026-08-13 found full key/row agreement for 1,776 learned summary rows and 144 jet summary rows, with maximum numeric differences of approximately $4.09\text{e-}14$ and $5.06\text{e-}14$, respectively [E15]. The supported statement is "numerically reproduced within floating-point precision."

A.5 Known output-isolation defect

The physics CLI changes working directory to the project root, so an isolated invocation refreshed ignored files under `projects/pnmf/outputs` before they were copied to `tmp/reason-report/physics-run/outputs` [E14, E17]. A reproducer must inspect

both locations until the command accepts an explicit output directory. This caveat concerns evidence custody and repeatability, not physical model accuracy.

A.6 Verified-scope reproduction record

The revised scope and physics evidence were generated into one isolated run directory:

```
projects/pnmf/.venv/Scripts/python.exe projects/pnmf/tools/run_verified_scope_validation.py `
--output-dir projects/pnmf/tmp/reason-report/verified_scope_20260813 `
--seed 20260724 --folds 3
```

The run manifest records the dirty Git state, datastore hash, official-list workbook hash, v6.3 aircraft/NPD hashes, registry version, exact eleven selected IDs, model parameters, software versions, calibration-artifact hash, and row counts. The principal artifacts are `verified_samples.csv`, `verified_group_splits.csv`, `merged_temporal_splits.csv`, `predictions.csv`, `summary.csv`, `verified_vs_merged_delta.csv`, `physics_validation.csv`, `physics_calibration_used.json`, and `run_manifest.json` [E19].

The release calibration candidate was built and compared with:

```
projects/pnmf/.venv/Scripts/python.exe projects/pnmf/pnmf_cli.py build-physics-calibration `
--output projects/pnmf/tmp/reason-report/verified_scope_20260813/physics_calibration_candidate.json `
--stage1-max-nfev 2 --joint-max-nfev 2 --spectral-grid 0,1.3,2.6
```

The candidate and release artifact are structurally identical in schema, anchor, source hashes, optimizer fields, parameters, and metrics; the release SHA-256 is recorded above. The candidate's in-sample metrics are 6.2007 dB RMSE, 5.3318 dB MAE, +0.4331 dB bias, and 9.8146 dB p90 absolute error over 180 cells. These are fit diagnostics, not external validation.

Appendix B. Evidence-bounded conclusion

PNMF is a traceable conceptual NPD screening framework with explicit separation of canonical truth, learned estimates, and a component-physics cross-check. The reviewed verified scope is now first-class: the versioned registry contains fourteen official-list families, exactly eleven trainable numerical v6.3 families, and three metadata-only families that remain unavailable for training. The learned API defaults to **verified**, while **merged** is retained as a labelled historical/research comparator. The model, CLI, Streamlit surface, storage identity, physics presets, and calibration artifact carry scope and provenance metadata [E19, E20].

Software and data correctness is **Verified** only where the fresh integrity checks and focused tests pass. Learned predictive validity is **Empirically supported within scope** for retrospective grouped results bounded to the eleven conventional verified families and the stated frozen reference/temporal protocols. The physics route provides an independent SEL/LAmax plausibility route, with a serialized A320-270N in-sample calibration and ten-family frozen validation stratified by estimates and fallbacks. Calibration fit is not external validation, and metadata-only verification does not create numerical training truth.

The following remains **Not established**: external accuracy, calibrated uncertainty, transfer to novel aircraft families or unconventional architectures, complete component-to-system external measurement validation, complete Doc 29 or ANOPP equivalence, certification suitability, and regulatory compliance. No predictive pass threshold is invented. Cross-tree dispersion and route deltas remain heuristics and diagnostics, not calibrated intervals; the historical component-physics timeout remains an execution-risk record even though the current full suite terminates.